

Analysis of the expected L_2 error of an over-parametrized deep neural network estimate learned by gradient descent without regularization

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Abstract: Recent results show that estimates defined by over-parametrized deep neural networks learned by applying gradient descent to a regularized empirical L_2 risk are universally consistent and achieve good rates of convergence. In this paper, we show that the regularization term is not necessary to obtain similar results. In the case of a suitably chosen initialization of the network, a suitable number of gradient descent steps, and a suitable step size we show that an estimate without a regularization term is universally consistent for bounded predictor variables. Additionally, we show that if the regression function is Hölder smooth with Hölder exponent $1/2 \leq p \leq 1$, the L_2 error converges to zero with a convergence rate of approximately $n^{-1/(1+d)}$. Furthermore, in case of an interaction model, where the regression function consists of a sum of Hölder smooth functions with d^* components, a rate of convergence is derived which does not depend on the input dimension d .

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1. Introduction

1.1. Deep learning

In the last decade, deep learning, i.e. the application of deep neural networks to data, has gained a lot of attraction in practical applications as well as in theoretical considerations and achieves impressive results. For instance, in natural science AlphaFold can predict protein structures (c.f. Billings et al. (2019)) or AlphaZero is able to outperform humans in three popular board games (c.f. McGrath et al. (2022)). Furthermore, the chatbot ChatGPT is one of the most capable language models nowadays, consisting of 175 billion parameters (c.f. Zong and Krishnamachari (2022)). Another example is the text-to-image model DALL-E-2, which consists of 3.5 billion parameters (c.f. Hunt (2023)). These observations suggest that a large number of parameters is very useful in practical applications of deep learning.

From a theoretical point of view this great success cannot yet be fully explained. However, there exist already some results concerning deep neural networks. For example, results on the rate of convergence of least squares estimates based on deep neural networks have been derived (cf. e.g., Bauer and Kohler (2019), Schmidt-Hieber (2020), Kohler and Langer (2021)).

Nevertheless, the above results neglect two features which have turned out to be very important for practical applications. First, in contrast to the results above, in practice estimates are computed using gradient descent instead of least squares. The second property they ignore is that practical applications often use over-parametrized neural networks. A network is called over-parametrized if the number of its parameters is much larger than the sample size. In this article, we specifically address the second issue by assuming that the number of weights grows polynomially with the sample size n . While many recent papers do not impose explicit restrictions on model size, our results show that even under a structured over-parametrization regime, universal consistency and good rates of convergence can still be achieved.

In this paper we consider a suitable over-parametrized deep neural network estimate computed by gradient descent and analyze its statistical performance in a nonparametric regression setting.

In addition to our theoretical analysis, we provide practical guidance on constructing over-parametrized neural networks that demonstrate strong empirical performance. In particular, we propose a specific number of gradient descent steps, an appropriate step size, and a degree of over-parametrization under which good generalization can still be achieved. This research offers insights that can inform best practices for applying over-parametrized models. Despite the large number of model parameters, our results indicate that reliable and consistent performance is achievable when the network architecture and training procedure are appropriately selected. This demonstrates that over-parametrization does not necessarily need to be avoided; in fact, it can be effectively managed. These results are particularly applicable to tasks such as image classification, natural language processing, and structured regression, where large neural network models are commonly used. Nonetheless, it must be noted that in specific domains, such as dynamic systems modeling, over-parametrized networks are not commonly used, and the assumptions of our theoretical framework may not fully reflect such settings.

1.2. Nonparametric regression

We analyze deep neural network estimates in the context of nonparametric regression. To do this we consider a random $\mathbb{R}^d \times \mathbb{R}$ -valued vector (X, Y) with $\mathbf{E}Y^2 < \infty$. Here we are interested in the dependence of the so-called *response variable* Y on the value of the *observation vector* X . We assume that it is possible to observe data of (X, Y) . This dataset is given by

$$\mathcal{D}_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$$

where $(X, Y), (X_1, Y_1), \dots, (X_n, Y_n)$ are independent and identically distributed (i.i.d.). Our aim is to construct an estimate $m_n(x) := m_n(x, \mathcal{D}_n)$ of the corresponding regression function $m : \mathbb{R}^d \rightarrow \mathbb{R}$ with $m(x) = \mathbf{E}\{Y|X = x\}$ such that the L_2 error

$$\int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx)$$

is small (cf. e.g., Györfi et al. (2002) for a detailed introduction to nonparametric regression).

There exist different modes of convergence. An important property an estimate should satisfy is to be universally consistent. Universal consistency means that the L_2 error converges to zero for all distributions of (X, Y) . But it does not provide any information about a rate of

convergence of the L_2 error. As shown in Theorem 7.2 and Problem 7.2 in [Devroye, Györfi and Lugosi \(1996\)](#) a rate of convergence cannot be derived in general since there always exists a distribution such that the L_2 error converges to zero arbitrarily slowly. Thus, in order to derive non-trivial results about the rate of convergence, we need to restrict the class of regression functions. To do this, we use the following definition of (p, C) -smoothness.

Definition 1.1. Let $p = q + s$ for some $q \in \mathbb{N}_0$ and $0 < s \leq 1$. A function $m : \mathbb{R}^d \rightarrow \mathbb{R}$ is called (p, C) -smooth, if for every $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$ with $\sum_{j=1}^d \alpha_j = q$ the partial derivative $\frac{\partial^q m}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}$ exists and satisfies

$$\left| \frac{\partial^q m}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}(x) - \frac{\partial^q m}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}(z) \right| \leq C \cdot \|x - z\|^s$$

for all $x, z \in \mathbb{R}^d$, where $\|\cdot\|$ denotes the Euclidean norm. For $p \leq 1$, the function m is called Hölder-smooth with exponent p and Hölder-constant C .

[Stone \(1982\)](#) showed that the optimal minimax rate of convergence in nonparametric regression for (p, C) -smooth functions is given by $n^{-2p/(2p+d)}$.

1.3. Main contributions

This paper makes several novel contributions to the statistical analysis of over-parametrized deep neural networks trained by gradient descent.

Firstly, we introduce a new estimate, defined as a linear combination of fully connected deep neural networks, which is trained with respect to the empirical L_2 risk without any explicit regularization term. This distinguishes our approach from existing methods that typically require regularization to ensure generalization.

Secondly, we present a new analytical framework for bounding the optimization error. In contrast to previous approaches that rely either on the Lipschitz continuity of the empirical L_2 risk (as in [Braun et al. \(2024\)](#)) or on its convexity (as in [Yehudai and Shamir \(2019\)](#)), our analysis combines both ideas in a unified argument, as formalized in Lemma 5.1.

Thirdly, the universal consistency of the proposed estimate is demonstrated under the mild condition of a bounded input support. This result indicates the efficacy of the proposed method in achieving asymptotically optimal performance under highly general conditions.

Fourthly, we derive a rate of convergence for the expected L_2 error. In the case where the regression function is (p, C) -smooth for $p \in [1/2, 1]$, the estimate achieves a rate of convergence close to

$$n^{-1/(1+d)}.$$

Furthermore, in the context of additive interaction models where each component function depends on at most d^* input variables, the rate improves to

$$n^{-1/(1+d^*)}$$

and becomes independent of the input dimension d .

The findings provide theoretical insights into the learning dynamics and statistical properties of over-parametrized networks. Specifically, they show that good performance can be achieved without explicit regularization.

1.4. Discussion of related results

From a theoretical point of view, there have been many interesting results in the past years. For instance, in the case of a suitably defined least squares estimate based on a multilayer neural network, [Kohler and Krzyżak \(2017\)](#) were able to derive a rate of convergence of $n^{-\frac{2p}{2p+d^*}}$ for (p, C) -smooth regression functions, which satisfy a model where the functions consist of a composition of functions applied to at most d^* components of its inputs, with $p \leq 1$. This rate does not depend on the input dimension d . [Bauer and Kohler \(2019\)](#) were able to show the same assertion for $p > 1$, provided that the activation function is sufficiently smooth. Under the additional condition of the so-called *sparsity* of the networks, [Schmidt-Hieber \(2020\)](#) showed the same rate of convergence for neural network estimates with ReLU activation function in case that the regression function satisfies a kind of hierarchical composition model. The property of sparsity is not necessary to obtain this rate of convergence for a regression function satisfying a hierarchical composition model ([Kohler and Langer \(2021\)](#)). [Suzuki \(2019\)](#) and [Suzuki and Nitanda \(2021\)](#) were able to show that this dimensional reduction can be achieved even under weaker assumptions on the smoothness of the regression function. For this purpose they considered deep learning with ReLU activation function for functions in Besov spaces.

In classical machine learning theory, the goal was to avoid over-parametrized neural networks. It was thought that these networks would lead to overfitting, where the weights fit the training data too closely and therefore generalize poorly to new data. Therefore, it was desired to obtain a bias-variance trade-off. The model should be complex enough to represent the structure of the underlying data, but simple enough to avoid overfitting. However, in practical applications, over-parametrized neural networks are often used very successfully and achieve high accuracy on new test data. This phenomenon has been studied recently by many different researchers (c.f., e.g. [Belkin et al. \(2019\)](#), [Frei, Chatterji and Bartlett \(2022\)](#) and the literature cited therein).

[Belkin et al. \(2019\)](#) were able to reconcile classical theory with new findings in practical applications and identified a pattern of performance dependence on unseen data and model capacity. This dependence is represented by the so-called *double descent curve*. It shows that if the model capacity exceeds the interpolation threshold, the test risk will decrease again.

In recent years, much work has focused on the capacity of neural networks. Researchers have tried to understand the ability of neural networks to adapt to the training data either on a finite data set ([Bubeck et al. \(2020\)](#)) or with respect to neural networks trained by gradient methods ([Daniely \(2019\)](#), [Daniely \(2020\)](#)) or in neural tangent training ([Montanari and Zhong \(2020\)](#)). In addition [Bartlett et al. \(2020\)](#) and [Belkin, Rakhlin and Tsybakov \(2019\)](#) have already shown that over-parametrized neural networks can achieve good rates of convergence.

In practical applications, the weights of neural network estimates are computed using gradient descent. For this reason, it is very interesting from a theoretical point of view to derive theoretical results for such estimates. [Nitanda and Suzuki \(2021\)](#) showed that the averaged stochastic gradient descent for over-parametrized neural networks with one hidden layer can achieve the optimal minimax rate of convergence in a neural tangent kernel setting where the smoothness of the regression function is measured by the neural tangent kernel. Furthermore, it was shown that the (stochastic) gradient descent can find a global minimum of the empirical L_2 risk for suitable over-parametrized deep neural networks ([Allen-Zhu, Li and Song \(2019\)](#), [Kawaguchi and Huang \(2019\)](#), [Arora et al. \(2019\)](#), [Du et al. \(2018\)](#)).

However, [Kohler and Krzyżak \(2021\)](#) were able to exhibit that over-parametrized deep neural networks minimizing the empirical L_2 risk do not generalize well on new data, in the sense that the networks that minimize the empirical risk do not achieve the optimal minimax rate of convergence.

[Braun et al. \(2024\)](#) showed that under the assumption that the Fourier transform of the regression function decays fast enough, a suitably initialized estimate learned by gradient descent achieves (up to a logarithmic factor) a rate of convergence of $n^{-\frac{1}{2}}$ which does not depend on the input dimension d . This result is related to a classical result of [Barron \(1994\)](#). He was able to derive a similar rate of convergence for a least squares neural network estimate in case that the Fourier transform has a finite first moment, which requires that the function becomes smoother with increasing dimension. [Kohler and Krzyżak \(2022a\)](#) analyzed the same estimates in an over-parametrized setting and were able to derive an improved rate of convergence of $n^{-\frac{2}{3}}$ for $d = 1$ using a suitably regularized L_2 risk.

The property of over-parametrization allows the gradient descent to find interpolating solutions that implicitly impose a regularization and leads to benign overfitting ([Bartlett, Montanari and Rakhlin \(2021\)](#)).

Many recent papers also consider minimax lower and upper L_2 error bounds for over-parametrized neural networks under increasingly smoothness or architectural assumptions ([Bartlett, Montanari and Rakhlin \(2021\)](#), and the references cited therein). These studies show that even highly expressive networks can achieve near-optimal rates of convergence under gradient-based training and illustrate the remarkable power of over-parametrization. However, most rely on either explicit regularization, particular kernel regimes, or tightly controlled network structures. In contrast, our analysis shows that simply minimizing the empirical L_2 risk without explicit regularization still yields universal consistency and good rates of convergence, providing a complementary perspective on the role of over-parametrization and implicit regularization in deep learning.

In the past years also lower bounds on the Minimax rate of convergence of regression estimates in case that the regression function is given by a (deep) neural network have been derived, see, e.g., [Klusowski and Barron \(2017\)](#), [Du, Wang and Zhai \(2018\)](#) and [Golestaneh, Taheri and Lederer \(2024\)](#) and the literature cited therein. E.g., [Golestaneh, Taheri and Lederer \(2024\)](#) shows that in case that the regression function is given by a suitable deep ReLU network, no estimate can achieve a rate of convergence faster than $1/\sqrt{n}$ (and not some rate of convergence approaching $1/n$ for smooth regression functions). This is in contrast to our work since we work in a different setup: we assume that the regression function satisfies some standard smoothness assumption and try to estimate it by deep neural networks, not that the regression function is a ReLU network and try to estimate this network (e.g., also by a deep neural network). In addition, corresponding upper bounds in these settings have been derived for least squares estimates and not for estimates learned by gradient descent as in our paper.

[Drews and Kohler \(2024\)](#) showed the property of universal consistency for over-parametrized deep neural network estimates computed by minimizing the L_2 risk with gradient descent. For these estimates, that are computed minimizing a regularized L_2 risk via gradient descent, [Kohler and Krzyżak \(2022b\)](#) derived a rate of convergence close to $n^{-\frac{1}{1+d}}$ for suitably smooth regression functions. For this, they considered three key ingredients. Firstly, they used that with high probability a subset of the inner initial weights has good properties. For a

suitable choice of the outer weights this causes a good approximation property. Secondly, for a suitably chosen number of gradient steps, a suitably chosen step size and suitably chosen bounds for the initial weights, they were able to use a metric entropy bound to control the generalization of the estimate. Thirdly, they analyzed the optimization of the empirical L_2 risk by optimizing the outer weights during the gradient descent and by using a regularization term.

Kohler and Krzyżak (2022b) were also able to show a dimension-independent rate of convergence for interaction models close to $n^{-\frac{1}{1+d^*}}$.

Our main results described in Subsection 1.3 are an extension of the results in Drews and Kohler (2024) and Kohler and Krzyżak (2022b). We show that the regularization term is not necessary. The estimate is universally consistent even by minimizing the empirical L_2 risk without a regularization term and achieves similar rates of convergence. As in both of the aforementioned articles, we assume that the regression function satisfies a Hölder smoothness condition which is a standard assumption in nonparametric regression. While the assumption may not in all real-world applications, such as in applications with highly irregular input-output relationships, it provides a useful starting point for theoretical analysis, and extending these results to more general settings remains an interesting direction for future work.

1.5. Limitations and future work

Our analysis is restricted to Hölder-smooth regression function with exponent $1/2$ and not to general (p, C) -smooth functions. Furthermore our estimates achieve a dimension reduction in case of interaction models but not in case of general hierarchical composition models. And finally we are able to analyze deep neural network with smooth activation function, but the case of the famous ReLU activation function is not covered. We hope that the analysis in this article is a first step towards more general results valid without the above limitations, and we are planning to adress these issues in future work. Also we hope that the derived theory will at some point give hints for modification of the definition of deep neural network estimates in practice (e.g., concerning the topology of the networks, the initialization of the weights, or details concerning gradient descent) which lead to estimates which achieve a better performance in practice.

1.6. Notation

Throughout this paper we will use the following notation: The sets of natural numbers, real numbers and non-negative real numbers are denoted by $\mathbb{N}$, $\mathbb{R}$ and $\mathbb{R}_+$, respectively. For $z \in \mathbb{R}$, we denote by $\lceil z \rceil$ the smallest integer greater than or equal to z and by $\lfloor z \rfloor$ the largest integer less than or equal to z . The Euclidean norm of $x \in \mathbb{R}^d$ is denoted by $\|x\|$. For a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, we refer to

$$\|f\|_\infty = \sup_{x \in \mathbb{R}^d} |f(x)|$$

as the supremum norm. Let $\mathcal{F}$ be a set of functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$, let $x_1, \dots, x_n \in \mathbb{R}^d$, set $x_1^n = (x_1, \dots, x_n)$ and let $p \geq 1$. A finite collection $f_1, \dots, f_N : \mathbb{R}^d \rightarrow \mathbb{R}$ is called an L_p

ε -cover of $\mathcal{F}$ on x_1^n if for any $f \in \mathcal{F}$ there exists $i \in \{1, \dots, N\}$ such that

$$\left(\frac{1}{n} \sum_{k=1}^n |f(x_k) - f_i(x_k)|^p \right)^{1/p} < \varepsilon.$$

The L_p ε -covering number of $\mathcal{F}$ on x_1^n is the size N of the smallest L_p ε -cover of $\mathcal{F}$ on x_1^n . It is denoted by $\mathcal{N}_p(\varepsilon, \mathcal{F}, x_1^n)$.

If A is a subset of $\mathbb{R}^d$ and $x \in \mathbb{R}^d$, then we set $1_A(x) = 1$ if $x \in A$ and $1_A(x) = 0$ otherwise.

For $z \in \mathbb{R}$ and $\beta > 0$ we define $T_\beta z = \max\{-\beta, \min\{\beta, z\}\}$. If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is a function and $\mathcal{F}$ is a set of such functions, then we set $(T_\beta f)(x) = T_\beta(f(x))$ and

$$T_\beta \mathcal{F} = \{T_\beta f : f \in \mathcal{F}\}.$$

1.7. Outline

In Section 2 we define the estimate. The main results concerning the universal consistency and the rate of convergence of the deep neural network estimate learned by gradient descent are presented in Section 3.

In Section 4 we illustrate the finite sample size performance of our estimate by applying it to simulated data, the proofs of the main result are presented in Section 5.

2. Definition of the estimate

In order to define the estimate, let $\sigma(x) = 1/(1 + e^{-x})$ be the logistic squasher. For the neural network topology, we choose a linear combination of K_n fully connected neural networks with L layers (including the output layer) and r neurons per layer. The output of the network is defined by

$$f_{\mathbf{w}}(x) = \sum_{k=1}^{K_n} w_{1,1,k}^{(L)} \cdot f_{k,1}^{(L)}(x) \quad (1)$$

for some $w_{1,1,1}^{(L)}, \dots, w_{1,1,K_n}^{(L)} \in \mathbb{R}$, where $f_{j,1}^{(L)}$ is recursively defined by

$$f_{k,i}^{(l)}(x) = \sigma \left(\sum_{j=1}^r w_{k,i,j}^{(l-1)} \cdot f_{k,j}^{(l-1)}(x) + w_{k,i,0}^{(l-1)} \right) \quad (2)$$

for some $w_{k,i,0}^{(l-1)}, \dots, w_{k,i,r}^{(l-1)} \in \mathbb{R}$ ($l = 2, \dots, L$) and

$$f_{k,i}^{(1)}(x) = \sigma \left(\sum_{j=1}^d w_{k,i,j}^{(0)} \cdot x^{(j)} + w_{k,i,0}^{(0)} \right) \quad (3)$$

for some $w_{k,i,0}^{(0)}, \dots, w_{k,i,d}^{(0)} \in \mathbb{R}$.

We denote by $f_{k,i}^{(l)}$ the output of neuron i in the l -th layer of the k -th fully connected network. By $w_{k,i,j}^{(l-1)}$ we denote the weight between neuron j in the $(l-1)$ -th layer and neuron i in the l -th layer of the k -th fully connected network.

The number of weights of the neural network is given by

$$W_n = K_n \cdot (1 + (r + 1) + (L - 2) \cdot r \cdot (r + 1) + r \cdot (d + 1)).$$

To obtain the estimate, we want to learn the weights using gradient descent. For this we initialize the weights $\mathbf{w}^{(0)} = ((\mathbf{w}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l}$ by setting

$$(\mathbf{w}^{(0)})_{1,1,k}^{(L)} = 0 \quad \text{for } k = 1, \dots, K_n, \quad (4)$$

and by choosing all components of $\mathbf{w}^{(0)}$ such that they are independent. We choose the weights $(\mathbf{w}^{(0)})_{k,i,j}^{(l)}$ with $l \in \{1, \dots, L - 1\}$ uniformly distributed on $[-20d \cdot (\log n)^2, 20d \cdot (\log n)^2]$ and $(\mathbf{w}^{(0)})_{k,i,j}^{(0)}$ uniformly distributed on $[-8d \cdot (\log n)^2 \cdot n^\tau, 8d \cdot (\log n)^2 \cdot n^\tau]$ for some fixed $\tau > 0$.

Set

$$\lambda_n = \frac{1}{t_n}$$

and compute

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \lambda_n \cdot (\nabla_{\mathbf{w}} F_n)(\mathbf{w}^{(t)})$$

for $t = 0, \dots, t_n - 1$ where

$$F_n(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}}(X_i) - Y_i|^2$$

is the empirical L_2 risk of the network $f_{\mathbf{w}}$ on the training data. The number of gradient descent steps t_n will be chosen in Section 3 below.

The estimate is then defined by

$$m_n(x) = T_{\beta_n} f_{\mathbf{w}^{(t_n)}}(x),$$

where $\beta_n = c_1 \cdot \log n$.

Because of (4) we have

$$F_n(\mathbf{w}^{(0)}) = \frac{1}{n} \sum_{i=1}^n |Y_i|^2.$$

3. Main results

3.1. Universal consistency

Our first result is the following theorem which presents the universal consistency for bounded X of an estimate learned by gradient descent using the empirical L_2 risk without a regularization term.

Theorem 3.1. *Let $\sigma(x) = 1/(1 + e^{-x})$ be the logistic squasher, and let $K_n, L, r \in \mathbb{N}$ and $\tau \in \mathbb{R}_+$. Furthermore, assume that $L \geq 2$, $r \geq 2d$, $\tau = 1/(d + 1)$,*

$$\frac{K_n}{n^\kappa} \rightarrow 0 \quad (n \rightarrow \infty) \quad (5)$$

for some $\kappa > 0$,

$$\frac{K_n}{n^{r+2}} \rightarrow \infty \quad (n \rightarrow \infty), \quad (6)$$

and $\beta_n = c_1 \cdot \log n$ for some $c_1 > 0$. Let the estimate m_n be defined as in Section 2 with

$$t_n = \lceil c_2 \cdot L_n \rceil \quad (7)$$

for some $c_2 \geq 1$ and $L_n > 0$ which satisfies

$$L_n \geq K_n^{3/2} \cdot (\log n)^{6L+5}.$$

Then we have

$$\mathbf{E} \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) \rightarrow 0 \quad (n \rightarrow \infty)$$

for every distribution of (X, Y) where $\text{supp}(X)$ is bounded and $\mathbf{E}Y^2 < \infty$.

Remark 1. The estimate is over-parametrized in the sense that the number of its parameters is much larger than the sample size. This is due to condition (6), which requires that K_n is asymptotically larger than $n^{r+2} \geq n^{2d+2}$. Theorem 3.1 shows that using the empirical L_2 risk without a regularization term, a suitable large number of steps, and a step size which is equal to the reciprocal of the number of steps provides a good generalization to new independent data.

Remark 2. The proof uses that the inner weights are chosen with high probability such that for properly chosen outer weights of the neural networks, the corresponding neural network has a small empirical L_2 risk. Hence, this neural network is based on representation guessing instead of representation learning.

3.2. Rate of convergence for (p, C) -smooth regression functions

Besides the result about the universal consistency of the estimate, it is interesting how fast the expected L_2 error converges to 0. Therefore, in the next theorem a rate of convergence for (p, C) -smooth functions is derived.

Theorem 3.2. Let $n \in \mathbb{N}$ and let $(X, Y), (X_1, Y_1), \dots, (X_n, Y_n)$ be independent and identically distributed $\mathbb{R}^d \times \mathbb{R}$ valued random variables which satisfy $\text{supp}(X) \subseteq [0, 1]^d$ and

$$\mathbf{E} \left\{ e^{c_3 \cdot Y^2} \right\} < \infty \quad (8)$$

for some $c_3 > 0$. Assume that the corresponding regression function $m(x) = \mathbf{E}\{Y|X = x\}$ is (p, C) -smooth for some $1/2 \leq p \leq 1$ and some $C > 0$.

Let $\sigma(x) = 1/(1 + e^{-x})$ be the logistic squasher, let $L, r \in \mathbb{N}$ with $L \geq 2$ and $r \geq 2d$. Set $\beta_n = c_1 \cdot \log n$ for some $c_1 > 0$,

$$K_n = n^{6d+r+2},$$

$$\tau = \frac{1}{1+d}.$$

Define the estimate m_n as in Section 2 with

$$t_n = \lceil c_4 \cdot L_n \rceil \quad (9)$$

for some $c_4 \geq 1$ and $L_n > 0$ which satisfies

$$L_n \geq K_n^{3/2} \cdot (\log n)^{6L+2}$$

and assume that

$$c_1 \cdot c_3 \geq 2. \quad (10)$$

Then we have for any $\epsilon > 0$

$$\mathbf{E} \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) \leq c_5 \cdot n^{-\frac{1}{1+d} + \epsilon}.$$

Remark 3. According to [Stone \(1982\)](#) the optimal minimax rate of convergence for (p, C) -smooth functions is $n^{-\frac{2p}{2p+d}}$. Thus, the rate of convergence derived from Theorem 3.2 is almost optimal for $p = \frac{1}{2}$. Unfortunately, if $p > \frac{1}{2}$, the rate of convergence derived above is not almost optimal. We assume that this is not a property of the estimate, but rather a consequence of our proof.

Remark 4. In our proofs the approximation error of the neural network does not improve if the smoothness p increases from $p = 1/2$ to some value greater $1/2$. Consequently, the convergence rate is not optimal when $p > 1/2$. To achieve better rates, an improved bound on the approximation error of the deep neural network is required.

3.3. Rate of convergence in an interaction model

The aim of this subsection is to modify the estimate defined above in order to obtain a rate of convergence that does not depend on d . For this we assume that the regression function satisfies

$$m(x) = \sum_{I \subseteq \{1, \dots, d\} : |I|=d^*} m_I(x_I),$$

with $1 \leq d^* < d$. The functions $m_I : \mathbb{R}^{|I|} \rightarrow \mathbb{R}$ ($I \subseteq \{1, \dots, d\}$, $|I| = d^*$) are (p, C) -smooth functions and we use the notation

$$x_I = (x^{(j_1)}, \dots, x^{(j_{d^*})})$$

for $I = \{j_1, \dots, j_{d^*}\}$.

The neural network is then given by

$$f_{\mathbf{w}}(x) = \sum_{I \subseteq \{1, \dots, d\} : |I|=d^*} f_{\mathbf{w}_I}(x_I)$$

where $f_{\mathbf{w}_I}$ is defined by (1)–(3) with d replaced by d^* and weight vector $\mathbf{w}_I$, where

$$\mathbf{w} = (\mathbf{w}_I)_{I \subseteq \{1, \dots, d\}, |I|=d^*}.$$

We initialize the weights $\mathbf{w}^{(0)} = \left(\left((\mathbf{w}_I^{(0)})^{(l)} \right)_{k,i,j,l} \right)_{I \subseteq \{1, \dots, d\}, |I|=d^*}$ by setting

$$(\mathbf{w}_I^{(0)})_{1,1,k}^{(L)} = 0 \quad (k = 1, \dots, K_n, I \subseteq \{1, \dots, d\}, |I| = d^*)$$

and by choosing all weights $(\mathbf{w}_I^{(0)})_{k,i,j}^{(l)}$ such that they are independent. We choose the weights $(\mathbf{w}_I^{(0)})_{k,i,j}^{(l)}$ with $l \in \{1, \dots, L-1\}$ uniformly distributed on the interval $[-20d^* \cdot (\log n)^2, 20d^* \cdot (\log n)^2]$ and we choose $(\mathbf{w}_I^{(0)})_{k,i,j}^{(0)}$ uniformly distributed on the interval $[-8d \cdot (\log n)^2 \cdot n^\tau, 8d \cdot (\log n)^2 \cdot n^\tau]$ where $\tau = 1/(1+d^*)$ ($I \subseteq \{1, \dots, d\}$ with $|I| = d^*$).

Similar as above we define the estimate as follows: Set

$$\lambda_n = \frac{1}{t_n}$$

and compute

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \lambda_n \cdot (\nabla_{\mathbf{w}} F_n)(\mathbf{w}^{(t)})$$

for $t = 0, \dots, t_n - 1$ where

$$F_n(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}}(X_i) - Y_i|^2$$

is the empirical L_2 risk of the network $f_{\mathbf{w}}$ on the training data. The number of gradient descent steps t_n will be chosen in Theorem 3.3 below.

The estimate is then defined by

$$m_n(x) = T_{\beta_n} f_{\mathbf{w}^{(t_n)}}(x)$$

where $\beta_n = c_1 \cdot \log n$.

Theorem 3.3. Let $d \in \mathbb{N}, d^* \in \{1, \dots, d\}, 1/2 \leq p \leq 1$. Furthermore, let $C > 0, n \in \mathbb{N}$ and let $(X, Y), (X_1, Y_1), \dots, (X_n, Y_n)$ be independent and identically distributed $\mathbb{R}^d \times \mathbb{R}$ valued random variables such that $\text{supp}(X) \subseteq [0, 1]^d$ and

$$\mathbf{E}\{e^{c_3 \cdot Y^2}\} < \infty$$

holds for some $c_3 > 0$. Assume that the corresponding regression function $m(x) = \mathbf{E}\{Y|X = x\}$ satisfies

$$m(x) = \sum_{I \subseteq \{1, \dots, d\}: |I|=d^*} m_I(x_I) \quad (x \in [0, 1]^d)$$

for some (p, C) -smooth functions $m_I : \mathbb{R}^{d^*} \rightarrow \mathbb{R}$ ($I \subseteq \{1, \dots, d\}, |I| = d^*$).

Let $\sigma(x) = 1/(1 + e^{-x})$ be the logistic squasher. Let $L, r \in \mathbb{N}$ with $L \geq 2$ and $r \geq 2d^*$. Set $\beta_n = c_1 \cdot \log n$,

$$K_n = n^{6d^*+r+2} \quad \text{and} \quad \tau = \frac{1}{1+d^*}.$$

Define the estimate m_n as in Subsection 3.3 with

$$t_n = \lceil c_6 \cdot L_n \rceil$$

for some $c_6 \geq 1$ and $L_n > 0$ which satisfies

$$L_n \geq K_n^{3/2} \cdot (\log n)^{6L+2}$$

and assume that

$$c_1 \cdot c_3 \geq 2. \quad (11)$$

Then we have for any $\epsilon > 0$

$$\mathbf{E} \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) \leq c_7 \cdot n^{-\frac{1}{1+d^*} + \epsilon}.$$

Remark 5. The optimal minimax rate of convergence $n^{-\frac{2p}{2p+d}}$ derived by [Stone \(1982\)](#) suffers from the curse of dimensionality. This means that if d is very large compared to p , the rate of convergence becomes extremely slow. However, [Stone \(1994\)](#) has already established that under appropriate assumptions it is possible to circumvent the curse of dimensionality. In Theorem 3.3, we were able to show that under the above assumptions on the regression function, the over-parametrized neural network estimate can also circumvent the curse of dimensionality.

4. Application to simulated data

In this section, we present the result of a simulation study to evaluate the performance of the over-parametrized neural network estimate learned by gradient descent introduced in Section 2. The objective of this section is to examine whether over-parametrization can, in practice, yield good generalization performance, contrary to the predictions of classical theory. We compare this estimate with a moderately over-parametrized standard fully connected neural network (OP-FC network), described in detail below.

The OP-FC network is defined analogously to the over-parametrized architecture in our results. Its output is given by

$$\bar{g}_w(x) = \sum_{k=1}^{\bar{r}} \bar{w}_{1,k}^{(\bar{L})} \cdot \bar{g}_k^{(\bar{L})}(x) \quad (12)$$

for some $\bar{w}_{1,1}^{(\bar{L})}, \dots, \bar{w}_{1,\bar{r}}^{(\bar{L})} \in \mathbb{R}$, where $\bar{g}_k^{(\bar{L})}$ is recursively defined by

$$\bar{g}_i^{(\bar{L})}(x) = \sigma \left(\sum_{j=1}^{\bar{r}} \bar{w}_{i,j}^{(\bar{L}-1)} \cdot \bar{g}_j^{(\bar{L}-1)}(x) + \bar{w}_{i,0}^{(\bar{L}-1)} \right) \quad (13)$$

for some $\bar{w}_{i,0}^{(\bar{L}-1)}, \dots, \bar{w}_{i,\bar{r}}^{(\bar{L}-1)} \in \mathbb{R}$ ($\bar{L} = 2, \dots, \bar{L}$) and

$$\bar{g}_i^{(1)}(x) = \sigma \left(\sum_{j=1}^d \bar{w}_{i,j}^{(0)} \cdot x^{(j)} + \bar{w}_{i,0}^{(0)} \right) \quad (14)$$

for some $\bar{w}_{i,0}^{(0)}, \dots, \bar{w}_{i,d}^{(0)} \in \mathbb{R}$.

The total number of trainable parameters is

$$\bar{W}_n = \bar{r} + (\bar{L} - 1) \cdot \bar{r} \cdot (\bar{r} + 1) + \bar{r} \cdot (d + 1).$$

We train the network with the PyTorch implementation of Adam, using $\bar{t}$ epochs and learning rate $\bar{\lambda}$.

Table 1

Hyper-parameter grids explored in the study. The learning rate and number of epochs are coupled for the over-parametrized model via $t_n = 1/\lambda_n$

Model	Grid values
Over-parametrized network	$K_n \in \{100, 200, 400, 600, 800\}$; $L_n \in \{4, 8, 16\}$ $r_n \in \{20, 40, 60\}$; $t_n \in \{300, 500, 700, 900\}$; $\lambda_n = 1/t_n$
OP-FC network	$\bar{L} \in \{4, 8, 16\}$; $\bar{r} \in \{20, 40, 60\}$; $\bar{t} \in \{300, 500, 700, 900\}$; $\bar{\lambda} \in \{0.001, 0.005, 0.01\}$

4.1. Simulation setup

The regression function employed in our simulations is the piecewise function $m : \mathbb{R} \rightarrow \mathbb{R}$ from Györfi et al. (2002), which is given by

$$m(x) = \begin{cases} \frac{(x+2)^2}{2} & \text{if } -1 \leq x < -0.5, \\ \frac{x}{2} + 0.875 & \text{if } -0.5 \leq x < 0, \\ -5 \cdot (x - 0.2)^2 + 1.075 & \text{if } 0 < x \leq 0.5, \\ x + 0.125 & \text{if } 0.5 \leq x < 1. \end{cases}$$

We draw

$$Y = m(X) + \sigma_N(X) \cdot N, \quad X \sim \mathcal{U}[-1, 1], \quad N \sim \mathcal{N}(0, 1),$$

with heteroscedastic noise scale $\sigma_N(x) = 0.2 - 0.1 \cos(2\pi x)$ and apply the estimates to samples of size $n = 1000$.

4.2. Training procedure and parameter selection

All networks are implemented in PyTorch. For the over-parametrized architecture we set the initial outer weights to $w_{1,1,k}^{(L)} = 0$ ($k = 1, \dots, K_n$). The initial weights of the first-layer, denoted by $w_{k,i,j}^{(0)}$, are drawn independently and uniformly from the interval $[-8d \cdot \log n, 8d \cdot \log n]$, and all remaining initial weights $w_{k,i,j}^{(l)}$ ($0 < l < L$) from the interval $[-20d \cdot (\log n)^2, 20d \cdot (\log n)^2]$.

We draw a sample of $n = 1000$ observations and first set aside $n_{\text{test}} = 500$ points as an independent test set. The remaining 500 observations form a development set. Since the performance of a neural-network is highly sensitive to hyper-parameters, we select them carefully. Therefore, we allocate $n_{\text{train}} = \lceil \frac{3}{4}n \rceil$ points within this development set for training and keep the remaining points as a validation set. Subsequently, a grid search is performed over the values of the hyper-parameters enumerated in Table 1. Each grid point (K_n, L_n, r_n, t_n) for the over-parametrized model and $(\bar{L}, \bar{r}, \bar{t}, \bar{\lambda})$ for the OP-FC model is trained ten times using independent random initializations of the data sets and the weights. The best hyper-parameter configurations are determined by the lowest empirical L_2 risk on the validation sets, averaged over all ten independent training runs.

The over-parametrized network is optimized by gradient descent using full-batch updates and mean-squared-error (MSE) loss, whereas the OP-FC network is trained with Adam also with a full-batch size. Furthermore, the weights of the OP-FC network estimate are initialized

by the Xavier initialization scheme while the weights of our over-parametrized network are initialized as described above. After selecting the top five hyper-parameter settings, we re-train each configuration 50 times, each time using newly generated random observations and an independent weight initialization. This procedure helps to avoid the risk that the results are overly influenced by a single favorable or unfavorable data realization. After training, the performance is evaluated on a separate test set of 500 unseen inputs. We report two metrics: test MSE and integrated L_2 risk.

4.3. Results

The mean values over 50 independent runs for the five best parameter combinations are summarized below in Table 2 and Table 3.

Table 2

Top five parameter combinations for the over-parametrized neural network ranked by lowest mean L_2 risk on the validation set over 50 runs

Model	Number of parameters	r_n	L_n	K_n	λ_n	t_n	MSE	Mean L_2 error
1	7969600	40	8	800	0.0011	900	0.048462	0.001681
2	721600	20	4	800	0.0011	900	0.049097	0.002215
3	2041200	40	4	600	0.0020	500	0.049140	0.001884
4	3565200	20	16	600	0.0020	500	0.050131	0.002793
5	4501200	60	4	600	0.0011	900	0.049338	0.001748

Table 3

Top five parameter combinations for the OP-FC network ranked by lowest mean L_2 risk on the validation set over 50 runs

Model	Number of parameters	$\bar{r}$	$\bar{L}$	$\bar{\lambda}$	$\bar{t}$	MSE	Mean L_2 error
1	25800	60	8	0.0050	900	0.049401	0.002334
2	11600	40	8	0.0050	900	0.051088	0.003479
3	11600	40	8	0.0100	900	0.049483	0.002910
4	25800	60	8	0.0100	700	0.051608	0.002889
5	25800	60	8	0.0100	900	0.049905	0.002667

The over-parametrized neural network with parameters $K_n = 800, L_n = 8, r_n = 40, t_n = 900$ achieves an MSE of 0.048462 and an L_2 error of 0.001681. In comparison with the OP-FC network (MSE = 0.049401, L_2 error = 0.002334), this model exhibits a reduction of approximately 1.9% in MSE and a reduction of approximately 28% in L_2 error.

Figure 1 displays the top five over-parametrized neural network estimates that are learned by gradient descent. Figure 2 shows the top five fully connected neural network estimates. The blue curves represent the true regression function $m(x)$, while the red curves illustrate the predictions of the representative model for each hyperparameter setting, selected based on the run with the closest L_2 error to the mean of the 50 runs. For each run, a different set of noisy data was generated as previously described. Overall the figures show that the over-parametrized network trained via gradient descent yields a lower mean L_2 error.

4.4. Discussion

The empirical results indicate that the over-parametrized network yields measurable but modest improvements over the best OP-FC network at the sample size $n = 1000$. For the

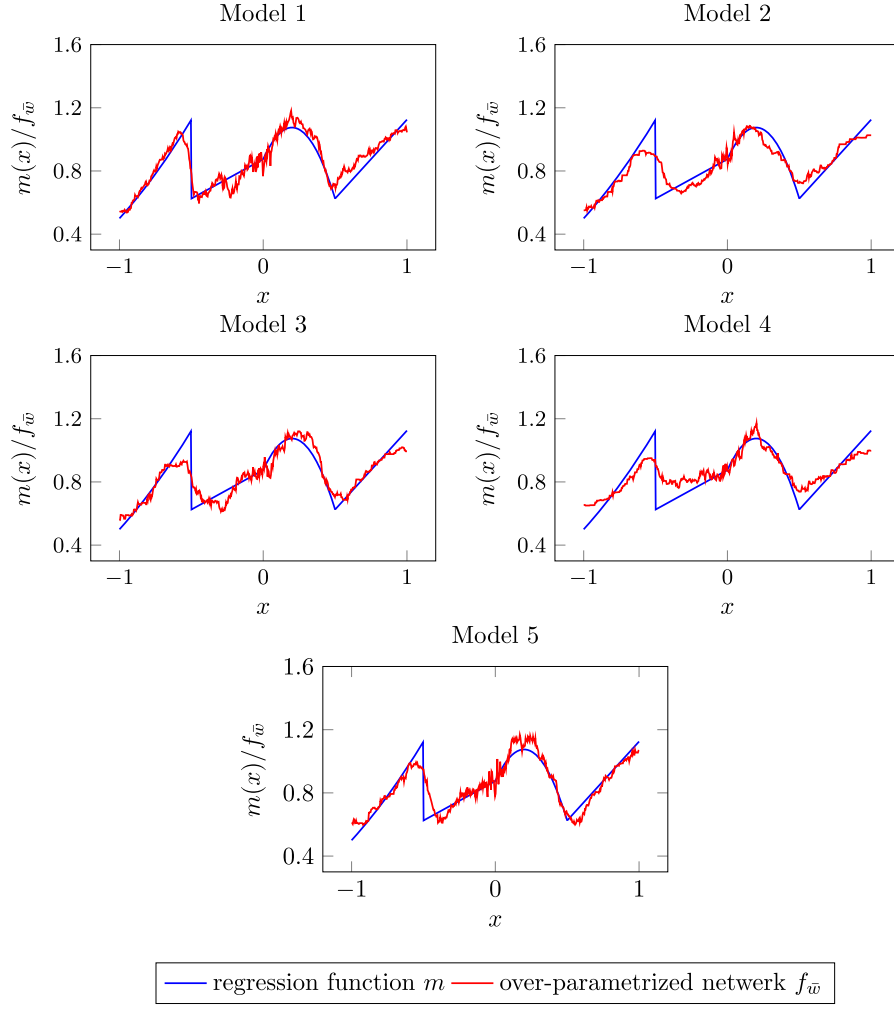


Fig 1. Visualization of the five best-performing over-parametrized models selected based on the mean validation L_2 error after hyper-parameter tuning.

most successful combination of the over-parametrized network $K_n = 800$, $L_n = 8$, $r_n = 40$, $t_n = 900$ the mean test MSE falls from 0.049401 to 0.048462 which is a relative drop of 1.9%. In contrast, the L_2 error decreases from 0.002334 to 0.001681, representing a notable 28% reduction. The difference between the two percentage reductions is consistent with the design of the regression function m . The test MSE evaluates pointwise accuracy on a random test set; because the noise variance is heteroscedastic and bounded below by $\sigma_N^2 \approx 0.01$, any improvement below about 10^{-3} is quickly masked by the irreducible noise, keeping the MSE improvement small. The L_2 error, by contrast, integrates the squared bias over the input space and is therefore more sensitive to systematic estimation errors. The observed 28% reduction may indicate that the over-parametrized network captures the shape of the regression function slightly better on average, even though improvements at individual points remain small.

In our opinion there are several reasons why our over-parametrized network trained by gradient descent achieves good results. On one hand gradient descent itself acts as an implicit regularizer. The very small MSE difference confirms that variance inflation is limited. This is consistent with the idea that in an over-parametrized network, gradient descent converges to

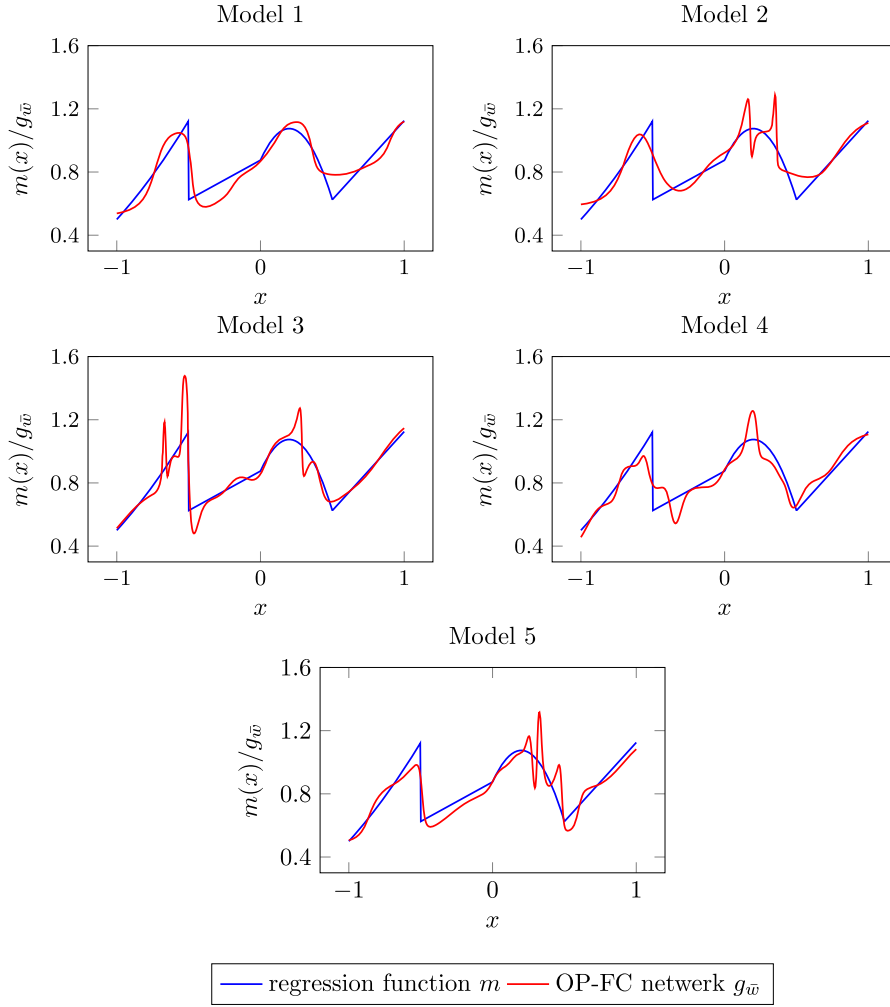


Fig 2. Visualization of the five best OP-FC models, selected based on the mean validation L_2 error after hyperparameter tuning.

the simplest (i.e., smoothest) model that still fits the training data perfectly (Gunasekar et al. (2018)).

On the other hand the given number of epochs t_n controls the model complexity. The coupling $\lambda_n = 1/t_n$ ties learning rate to epoch count which reduces the complexity of deep neural networks. This reduction in complexity is achieved by preventing the network from deviating into solutions with high variance.

In the results there are noticeable fluctuations in the errors. These are due to the randomness of the data, which results in randomness of the parameter combinations chosen and the estimation of the errors of the corresponding estimates. However, across the five retained overparameterized settings, the mean L_2 error reduction ranged from 4% to 36%, ruling out a single lucky hyperparameter.

The experiment is one-dimensional, and therefore the scaling of the variance term when d is large is not revealed. The extension of the study to higher-dimensional spaces is considered as an extension of the study.

In summary, for an over-parametrized network trained by simple gradient descent, the

integrated bias is reduced with a negligible variance penalty when $n = 1000$. The findings of the present study indicate that over-parametrization need not be avoided when combined with implicit regularization by the optimizer. The analysis highlights the potential performance benefits of over-parametrization combined with gradient descent in nonparametric regression.

5. Proofs

5.1. Auxiliary results for the proof of Theorem 3.1

In this section we will present auxiliary results that are necessary for the proof of Theorem 3.1. The first auxiliary result enables us to analyze the gradient descent.

Lemma 5.1. *Let $d_1, d_2 \in \mathbb{N}$, let $(u_0, v_0) \in \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$, let $F : \mathbb{R}^{d_1} \times \mathbb{R}^{d_2} \rightarrow \mathbb{R}_+$ be a continuously differentiable function and set*

$$A = \left\{ (u, v) \in \mathbb{R}^{d_1} \times \mathbb{R}^{d_2} : \|(u, v) - (u_0, v_0)\| \leq 2 \cdot \sqrt{F(u_0, v_0)} + 1 \right\}.$$

Let $u^* \in \mathbb{R}^{d_1}$ and assume that for some $D_n, L_n > 0$ it holds

$$u \mapsto F(u, v) \quad \text{is convex for all } v \in \mathbb{R}^{d_2},$$

$$\|(\nabla_{(u,v)} F)(u, v)\| \leq L_n \tag{15}$$

for all $(u, v) \in A$,

$$\|(\nabla_{(u,v)} F)(u_1, v_1) - (\nabla_{(u,v)} F)(u_2, v_2)\| \leq L_n \cdot \|(u_1, v_1) - (u_2, v_2)\| \tag{16}$$

for all $(u_1, v_1), (u_2, v_2) \in A$ and

$$|F(u^*, v) - F(u^*, v_0)| \leq D_n \cdot \|u^*\| \cdot \|v - v_0\| \tag{17}$$

for all $v \in \{\tilde{v} : \|\tilde{v} - v_0\| \leq \sqrt{2 \cdot F(u_0, v_0)}\}$.

Set

$$u_{t+1} = u_t - \lambda \cdot (\nabla_u F)(u_t, v_t),$$

$$v_{t+1} = v_t - \lambda \cdot (\nabla_v F)(u_t, v_t)$$

for $t = 0, 1, \dots, t_n - 1$, where

$$t_n \geq L_n \quad \text{and} \quad \lambda = \frac{1}{t_n}.$$

Then we have

$$F(u_{t_n}, v_{t_n}) \leq F(u^*, v_0) + D_n \cdot \|u^*\| \cdot \sqrt{2 \cdot F(u_0, v_0)} + \frac{\|u^* - u_0\|^2}{2} + \frac{F(u_0, v_0)}{t_n}.$$

Furthermore, even if (17) does not hold, we have

$$\|u_t - u_0\| \leq \sqrt{2 \cdot F(u_0, v_0)} \quad \text{and} \quad \|v_t - v_0\| \leq \sqrt{2 \cdot F(u_0, v_0)}$$

for all $t = 0, 1, \dots, t_n$.

Proof. In the first step of the proof we show

$$\frac{1}{t_n} \sum_{t=0}^{t_n-1} F(u_t, v_t) \leq \frac{1}{t_n} \sum_{t=0}^{t_n-1} F(u^*, v_t) + \frac{\|u^* - u_0\|^2}{2} + \frac{1}{2 \cdot t_n} \sum_{t=0}^{t_n-1} \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2. \quad (18)$$

By convexity of $u \mapsto F(u, v_t)$ we have

$$\begin{aligned} & F(u_t, v_t) - F(u^*, v_t) \\ & \leq \langle (\nabla_u F)(u_t, v_t), u_t - u^* \rangle \\ & = \frac{1}{2 \cdot \lambda} \cdot 2 \cdot \langle \lambda \cdot (\nabla_u F)(u_t, v_t), u_t - u^* \rangle \\ & \leq \frac{1}{2 \cdot \lambda} \cdot \left(-\|u_t - u^* - \lambda \cdot (\nabla_u F)(u_t, v_t)\|^2 + \|u_t - u^*\|^2 + \|\lambda \cdot (\nabla_u F)(u_t, v_t)\|^2 \right) \\ & = \frac{1}{2 \cdot \lambda} \cdot \left(\|u_t - u^*\|^2 - \|u_{t+1} - u^*\|^2 + \lambda^2 \cdot \|(\nabla_u F)(u_t, v_t)\|^2 \right). \end{aligned}$$

This implies

$$\begin{aligned} & \frac{1}{t_n} \sum_{t=0}^{t_n-1} F(u_t, v_t) - \frac{1}{t_n} \sum_{t=0}^{t_n-1} F(u^*, v_t) \\ & = \frac{1}{t_n} \sum_{t=0}^{t_n-1} (F(u_t, v_t) - F(u^*, v_t)) \\ & \leq \frac{1}{t_n} \sum_{t=0}^{t_n-1} \frac{1}{2 \cdot \lambda} \cdot \left(\|u_t - u^*\|^2 - \|u_{t+1} - u^*\|^2 \right) + \frac{1}{t_n} \sum_{t=0}^{t_n-1} \frac{\lambda}{2} \cdot \|(\nabla_u F)(u_t, v_t)\|^2 \\ & = \frac{1}{2} \cdot \sum_{t=0}^{t_n-1} \left(\|u_t - u^*\|^2 - \|u_{t+1} - u^*\|^2 \right) + \frac{1}{2 \cdot t_n} \sum_{t=0}^{t_n-1} \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2 \\ & \leq \frac{\|u_0 - u^*\|^2}{2} + \frac{1}{2 \cdot t_n} \sum_{t=0}^{t_n-1} \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2. \end{aligned}$$

In the second step of the proof we show that

$$\begin{aligned} & \|(\nabla_{(u,v)} F)((u_t, v_t) + \tau \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t))) - (\nabla_{(u,v)} F)(u_t, v_t)\| \\ & \leq L_n \cdot \tau \cdot \|(u_{t+1}, v_{t+1}) - (u_t, v_t)\| \end{aligned} \quad (19)$$

for all $\tau \in [0, 1]$ implies

$$F(u_{t+1}, v_{t+1}) - F(u_t, v_t) \leq -\frac{1}{2} \cdot \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2 - \frac{1}{2} \cdot \lambda \cdot \|(\nabla_v F)(u_t, v_t)\|^2. \quad (20)$$

The function

$$H : [0, 1] \rightarrow \mathbb{R}, \quad H(\tau) = F((u_t, v_t) + \tau \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t)))$$

is continuously differentiable. By the fundamental theorem of calculus, assumption (19) and $\lambda \leq 1/L_n$ we get

$$F(u_{t+1}, v_{t+1}) - F(u_t, v_t) = H(1) - H(0) = \int_0^1 H'(\tau) d\tau$$

$$\begin{aligned}
&= \int_0^1 (\nabla_{(u,v)} F)((u_t, v_t) + \tau \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t))) \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t)) d\tau \\
&= \int_0^1 ((\nabla_{(u,v)} F)((u_t, v_t) + \tau \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t))) - (\nabla_{(u,v)} F)(u_t, v_t)) \\
&\quad \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t)) d\tau \\
&\quad + \int_0^1 (\nabla_{(u,v)} F)(u_t, v_t) \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t)) d\tau \\
&\leq \int_0^1 \|(\nabla_{(u,v)} F)((u_t, v_t) + \tau \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t))) - (\nabla_{(u,v)} F)(u_t, v_t)\| \\
&\quad \cdot \|(u_{t+1}, v_{t+1}) - (u_t, v_t)\| d\tau \\
&\quad + (\nabla_{(u,v)} F)(u_t, v_t) \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t)) \\
&\leq \int_0^1 L_n \cdot \tau \cdot \|(u_{t+1}, v_{t+1}) - (u_t, v_t)\|^2 d\tau \\
&\quad + (\nabla_{(u,v)} F)(u_t, v_t) \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t)) \\
&= \frac{1}{2} \cdot L_n \cdot \|(u_{t+1}, v_{t+1}) - (u_t, v_t)\|^2 + (\nabla_{(u,v)} F)(u_t, v_t) \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t)) \\
&= \frac{1}{2} \cdot L_n \cdot \left(\lambda^2 \cdot \|(\nabla_u F)(u_t, v_t)\|^2 + \lambda^2 \cdot \|(\nabla_v F)(u_t, v_t)\|^2 \right) - \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2 \\
&\quad - \lambda \cdot \|(\nabla_v F)(u_t, v_t)\|^2 \\
&= \lambda \cdot \left(\frac{1}{2} \cdot L_n \cdot \lambda - 1 \right) \cdot \|(\nabla_u F)(u_t, v_t)\|^2 + \lambda \cdot \left(\frac{1}{2} \cdot L_n \cdot \lambda - 1 \right) \cdot \|(\nabla_v F)(u_t, v_t)\|^2 \\
&\leq -\frac{1}{2} \cdot \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2 - \frac{1}{2} \cdot \lambda \cdot \|(\nabla_v F)(u_t, v_t)\|^2.
\end{aligned}$$

In the *third step of the proof* we show

$$F(u_1, v_1) - F(u_0, v_0) \leq -\frac{1}{2} \cdot \lambda \cdot \|(\nabla_u F)(u_0, v_0)\|^2 - \frac{1}{2} \cdot \lambda \cdot \|(\nabla_v F)(u_0, v_0)\|^2.$$

By (15) we know

$$\|(\nabla_{(u,v)} F)(u_0, v_0)\| \leq L_n,$$

which implies for any $\tau \in [0, 1]$

$$\|(u_0, v_0) + \tau \cdot ((u_1, v_1) - (u_0, v_0)) - (u_0, v_0)\| \leq \lambda \cdot L_n.$$

Consequently we can conclude from (16) that (19) holds for $t = 0$, from which we get the assertion of step 3 by applying the result from step 2.

In the *fourth step of the proof* we show that by induction on t , that

$$F(u_{t+1}, v_{t+1}) - F(u_t, v_t) \leq -\frac{1}{2} \cdot \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2 - \frac{1}{2} \cdot \lambda \cdot \|(\nabla_v F)(u_t, v_t)\|^2$$

holds for all $t \in \{0, 1, \dots, t_n - 1\}$, and that

$$\|u_t - u_0\| \leq \sqrt{2 \cdot F(u_0, v_0)} \quad \text{and} \quad \|v_t - v_0\| \leq \sqrt{2 \cdot F(u_0, v_0)}$$

hold for all $t \in \{0, 1, \dots, t_n\}$.

For $t = 0$ the assertion follows from step 3. So assume now that the assertion holds for some $t \in \{0, 1, \dots, t_n - 1\}$. Then (15) implies

$$\|(\nabla_{(u,v)} F)(u_t, v_t)\| \leq L_n,$$

which, for any $\tau \in [0, 1]$, results in

$$\begin{aligned} \|(u_t, v_t) + \tau \cdot ((u_{t+1}, v_{t+1}) - (u_t, v_t)) - (u_0, v_0)\| &\leq \|(u_t, v_t) - (u_0, v_0)\| + \lambda \cdot L_n \\ &\leq \sqrt{\|u_t - u_0\|^2 + \|v_t - v_0\|^2} + \lambda \cdot L_n \leq 2 \cdot \sqrt{F(u_0, v_0)} + \lambda \cdot L_n. \end{aligned}$$

Consequently, we can conclude from inequality (16) that (19) holds, from which we get by step 2

$$F(u_{t+1}, v_{t+1}) - F(u_t, v_t) \leq -\frac{1}{2} \cdot \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2 - \frac{1}{2} \cdot \lambda \cdot \|(\nabla_v F)(u_t, v_t)\|^2.$$

Furthermore, we have

$$\begin{aligned} &\|u_{t+1} - u_0\| \\ &\leq \sum_{s=0}^t \|u_{s+1} - u_s\| \\ &\leq \sqrt{(t+1) \cdot \sum_{s=0}^t \|u_{s+1} - u_s\|^2} \\ &= \sqrt{(t+1) \cdot \sum_{s=0}^t \lambda^2 \cdot \|(\nabla_u F)(u_s, v_s)\|^2} \\ &\leq \sqrt{2 \cdot (t+1) \cdot \lambda \cdot \sum_{s=0}^t (F(u_s, v_s) - F(u_{s+1}, v_{s+1}))} \\ &\leq \sqrt{2 \cdot (t+1) \cdot \frac{1}{t_n} \cdot F(u_0, v_0)} \\ &\leq \sqrt{2 \cdot F(u_0, v_0)} \end{aligned}$$

and

$$\begin{aligned} &\|v_{t+1} - v_0\| \\ &\leq \sum_{s=0}^t \|v_{s+1} - v_s\| \\ &\leq \sqrt{(t+1) \cdot \sum_{s=0}^t \|v_{s+1} - v_s\|^2} \\ &= \sqrt{(t+1) \cdot \sum_{s=0}^t \lambda^2 \cdot \|(\nabla_v F)(u_s, v_s)\|^2} \\ &\leq \sqrt{2 \cdot (t+1) \cdot \lambda \cdot \sum_{s=0}^t (F(u_s, v_s) - F(u_{s+1}, v_{s+1}))} \end{aligned}$$

$$\begin{aligned}
&\leq \sqrt{2 \cdot (t+1) \cdot \frac{1}{t_n} \cdot F(u_0, v_0)} \\
&\leq \sqrt{2 \cdot F(u_0, v_0)}.
\end{aligned}$$

In the *fifth step of the proof* we show the assertion of Lemma 5.1. By the fourth step we know that F is monotonically decreasing hence it holds

$$F(u_{t_n}, v_{t_n}) \leq \frac{1}{t_n} \sum_{t=0}^{t_n-1} F(u_t, v_t).$$

Then we can conclude from the first step of the proof

$$\begin{aligned}
&F(u_{t_n}, v_{t_n}) \\
&\leq \frac{1}{t_n} \sum_{t=0}^{t_n-1} F(u^*, v_t) + \frac{\|u^* - u_0\|^2}{2} + \frac{1}{2 \cdot t_n} \sum_{t=0}^{t_n-1} \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2 \\
&\leq F(u^*, v_0) + \frac{1}{t_n} \sum_{t=0}^{t_n-1} |F(u^*, v_t) - F(u^*, v_0)| \\
&\quad + \frac{\|u^* - u_0\|^2}{2} + \frac{1}{2 \cdot t_n} \sum_{t=0}^{t_n-1} \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2.
\end{aligned}$$

By assumption (17) and the fourth step of the proof we get

$$\frac{1}{t_n} \sum_{t=0}^{t_n-1} |F(u^*, v_t) - F(u^*, v_0)| \leq \frac{1}{t_n} \sum_{t=0}^{t_n-1} D_n \cdot \|u^*\| \cdot \|v_t - v_0\| \leq D_n \cdot \|u^*\| \cdot \sqrt{2 \cdot F(u_0, v_0)}.$$

As in the fourth step of the proof we get

$$\sum_{t=0}^{t_n-1} \lambda \cdot \|(\nabla_u F)(u_t, v_t)\|^2 \leq 2 \cdot \sum_{t=0}^{t_n-1} (F(u_t, v_t) - F(u_{t+1}, v_{t+1})) \leq 2 \cdot F(u_0, v_0).$$

Summarizing the above results, the proof is complete. $\square$

With the following two results, we can show that the assumptions (15) and (16) are satisfied in the proof of Theorem 3.1.

Lemma 5.2. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be bounded and differentiable, and assume that its derivative is bounded. Let $\alpha_n \geq 1$, $t_n \geq L_n$, $\gamma_n^* \geq 1$, $B_n \geq 1$, $r \geq 2d$,*

$$|w_{1,1,k}^{(L)}| \leq \gamma_n^* \quad \text{for } k = 1, \dots, K_n, \quad (21)$$

$$|w_{k,i,j}^{(l)}| \leq B_n \quad \text{for } l = 1, \dots, L-1 \quad (22)$$

and

$$\|\mathbf{w} - \mathbf{v}\|_\infty^2 \leq \frac{8t_n}{L_n} \cdot \max\{F_n(\mathbf{v}), 1\}. \quad (23)$$

Then we have for $X_1, \dots, X_n \in [-\alpha_n, \alpha_n]^d$

$$\|(\nabla_{\mathbf{w}} F_n)(\mathbf{w})\| \leq c_8 \cdot K_n^{3/2} \cdot B_n^{2L} \cdot (\gamma_n^*)^2 \cdot \alpha_n^2 \cdot \sqrt{\frac{t_n}{L_n} \cdot \max\{F_n(\mathbf{v}), 1\}}.$$

Proof. The proof follows from Lemma 2 in [Drews and Kohler \(2024\)](#). For sake of completeness the proof is given in the appendix. $\square$

Lemma 5.3. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be bounded and differentiable, and assume that its derivative is Lipschitz continuous and bounded. Let $\alpha_n \geq 1$, $t_n \geq L_n$, $\gamma_n^* \geq 1$, $B_n \geq 1$, $r \geq 2d$ and assume*

$$\left| \max\{(\mathbf{w}_1)_{1,1,k}^{(L)}, (\mathbf{w}_2)_{1,1,k}^{(L)}\} \right| \leq \gamma_n^* \quad \text{for } k = 1, \dots, K_n, \quad (24)$$

$$\left| \max\{(\mathbf{w}_1)_{k,i,j}^{(l)}, (\mathbf{w}_2)_{k,i,j}^{(l)}\} \right| \leq B_n \quad \text{for } l = 1, \dots, L-1 \quad (25)$$

and

$$\|\mathbf{w}_2 - \mathbf{v}\|^2 \leq 8 \cdot \frac{t_n}{L_n} \cdot \max\{F_n(\mathbf{v}), 1\}. \quad (26)$$

Then we have for $X_1, \dots, X_n \in [-\alpha_n, \alpha_n]^d$

$$\begin{aligned} & \|(\nabla_{\mathbf{w}} F_n)(\mathbf{w}_1) - (\nabla_{\mathbf{w}} F_n)(\mathbf{w}_2)\| \\ & \leq c_9 \cdot \max\{\sqrt{F_n(\mathbf{v})}, 1\} \cdot (\gamma_n^*)^2 \cdot B_n^{3L} \cdot \alpha_n^3 \cdot K_n^{3/2} \cdot \sqrt{\frac{t_n}{L_n}} \cdot \|\mathbf{w}_1 - \mathbf{w}_2\|. \end{aligned}$$

Proof. The proof follows from Lemma 3 in [Drews and Kohler \(2024\)](#). For the sake of completeness the complete proof is given in the appendix. $\square$

The next auxiliary result uses a metric entropy bound to control the complexity of a set of over-parametrized deep neural networks.

Lemma 5.4. *Let $\alpha \geq 1$, $\beta > 0$ and let $A, B, C \geq 1$. Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be k -times differentiable such that all derivatives up to order k are bounded on $\mathbb{R}$. Let $\mathcal{F}$ be the set of all functions $f_{\mathbf{w}}$ defined by (1)–(3) where the weight vector $\mathbf{w}$ satisfies*

$$\sum_{j=1}^{K_n} |w_{1,1,j}^{(L)}| \leq C, \quad (27)$$

$$|w_{k,i,j}^{(l)}| \leq B \quad (k \in \{1, \dots, K_n\}, i, j \in \{1, \dots, r\}, l \in \{1, \dots, L-1\}) \quad (28)$$

and

$$|w_{k,i,j}^{(0)}| \leq A \quad (k \in \{1, \dots, K_n\}, i \in \{1, \dots, r\}, j \in \{1, \dots, d\}). \quad (29)$$

Then we have for any $1 \leq p < \infty$, $0 < \epsilon < \beta$ and $x_1^n \in (\mathbb{R}^d)^n$

$$\begin{aligned} & \mathcal{N}_p \left(\epsilon, \{T_{\beta} f \cdot 1_{[-\alpha, \alpha]^d} : f \in \mathcal{F}\}, x_1^n \right) \\ & \leq \left(c_{10} \cdot \frac{\beta^p}{\epsilon^p} \right)^{c_{11} \cdot \alpha^d \cdot B^{(L-1) \cdot d} \cdot A^d \cdot \left(\frac{C}{\epsilon}\right)^{d/k} + c_{12}}. \end{aligned}$$

Proof. See Lemma 4 in [Drews and Kohler \(2024\)](#). $\square$

The following lemma provides a bound on the error of approximating a Lipschitz continuous and bounded function using an over-parametrized deep neural network.

Lemma 5.5. Let σ be the logistic squasher, let $1 \leq \alpha_n \leq \log n$, let $m : \mathbb{R}^d \rightarrow \mathbb{R}$ be Lipschitz continuous as well as bounded, let $L, r, n \in \mathbb{N}$ with $L \geq 2$, $r \geq 2d$, $n \geq 8d$ and $n \geq \exp(r+1)$ and let $K, N_n \in \mathbb{N}$ with $2 \leq K \leq \alpha_n - 1$ and $N_n \cdot (K^2 + 1)^d \leq K_n$. Given $u_1, v_1, \dots, u_{N_n(K^2+1)^d}, v_{N_n(K^2+1)^d} \in [-K - \frac{2}{K}, K]^d$, choose $\mathbf{w}$ such that

$$w_{k,j,j}^{(0)} = 4d \cdot K^2 \cdot (\log n)^2 \quad \text{and} \quad w_{k,j,0}^{(0)} = -4d \cdot K^2 \cdot (\log n)^2 \cdot u_k^{(j)} \quad \text{for } j \in \{1, \dots, d\}, \quad (30)$$

$$w_{k,j+d,j}^{(0)} = -4d \cdot K^2 \cdot (\log n)^2 \quad \text{and} \quad w_{k,j+d,0}^{(0)} = 4d \cdot K^2 \cdot (\log n)^2 \cdot v_k^{(j)} \quad \text{for } j \in \{1, \dots, d\}, \quad (31)$$

$$w_{k,s,t}^{(0)} = 0 \quad \text{if } s \leq 2d, s \neq t, s \neq t+d \text{ and } t > 0, \quad (32)$$

$$w_{k,1,t}^{(1)} = 8 \cdot (\log n)^2 \quad \text{for } t \in \{1, \dots, 2d\}, \quad (33)$$

$$w_{k,1,0}^{(1)} = -8 \cdot (\log n)^2 \left(2d - \frac{1}{n} \right), \quad (34)$$

$$w_{k,1,t}^{(1)} = 0 \quad \text{for } t > 2d, \quad (35)$$

$$w_{k,1,l}^{(l)} = 6 \cdot (\log n)^2 \quad \text{for } l \in \{2, \dots, L-1\}, \quad (36)$$

$$w_{k,1,0}^{(l)} = -3 \cdot (\log n)^2 \quad \text{for } l \in \{2, \dots, L-1\} \quad (37)$$

and

$$w_{k,1,t}^{(l)} = 0 \quad \text{for } t > 1 \text{ and } l \in \{2, \dots, L-1\} \quad (38)$$

for all $k \in \{1, \dots, N_n \cdot (K^2 + 1)^d\}$.

Then there exist $u_1, v_1, \dots, u_{N_n(K^2+1)^d}, v_{N_n(K^2+1)^d} \in [-K - \frac{2}{K}, K]^d$ and

$$\alpha_1, \dots, \alpha_{N_n(K^2+1)^d} \in \left[-\frac{\|m\|_\infty}{N_n}, \frac{\|m\|_\infty}{N_n} \right] \quad (39)$$

such that for all pairwise distinct $j_1, \dots, j_{N_n(K^2+1)^d} \in \{1, \dots, K_n\}$

$$\begin{aligned} & \int |f_{\bar{\mathbf{w}}}(x) - m(x)|^2 \mathbf{P}_X(dx) \\ & \leq c_{13} \cdot \left(\frac{1}{K} + \frac{N_n^2 \cdot K^{4d}}{n^2} + \left(\frac{K^{2d}}{n} + 1 \right)^2 \cdot \mathbf{P}_X(\mathbb{R}^d \setminus [-K, K]^d) \right) \end{aligned} \quad (40)$$

holds for all weight vectors $\bar{\mathbf{w}}$ which satisfy

$$\bar{w}_{1,1,j_k}^{(L)} = \alpha_k \quad (k \in \{1, \dots, N_n \cdot (K^2 + 1)^d\}), \quad \bar{w}_{1,1,k}^{(L)} = 0 \quad (k \notin \{j_1, \dots, j_{N_n(K^2+1)^d}\}) \quad (41)$$

and

$$\left| w_{s,k,i}^{(l)} - \bar{w}_{j_s,k,i}^{(l)} \right| \leq \log n \quad \text{for all } l \in \{0, \dots, L-1\}, s \in \{1, \dots, N_n \cdot (K^2 + 1)^d\}. \quad (42)$$

Additionally we get

$$\|f_{\bar{\mathbf{w}}}\|_\infty \leq c_{14} \cdot \left(3^d + \frac{(K^2 + 1)^d}{n} \right) \quad (43)$$

where c_{14} depends on $\|m\|_\infty$.

Proof. Subdivide the cube $[-K - \frac{2}{K}, K]^d$ in $(K^2 + 1)^d$ equidistant cubes C_i of side length $\frac{2}{K}$. For simplicity we number these cubes C_i by $i \in \{1, \dots, (K^2 + 1)^d\}$, such that C_i corresponds to the cube

$$[u_i^{(1)}, v_i^{(1)}] \times \dots \times [u_i^{(d)}, v_i^{(d)}].$$

Let C_{Lip} be the Lipschitz constant of m .

We apply Lemma 6 from [Drews and Kohler \(2024\)](#) with $a_i = -K - \frac{2}{K}$, $b_i = K$ and $K^2 + 1$ instead of K to m/N_n and $\delta = \frac{1}{K^2}$. This results in

$$\left| f_{\tilde{w}}(x) - \frac{1}{N_n} \cdot m(x) \right| \leq c_{15} \cdot \left(\frac{C_{Lip}}{N_n} \cdot \frac{2}{K} + (K^2 + 1)^d \cdot \frac{1}{n} \right)$$

for all $x \in [-K - \frac{2}{K}, K]^d$ which are not contained in

$$A := \bigcup_{i \in \{0, 1, \dots, K^2 + 1\}} \bigcup_{j \in \{1, \dots, d\}} \left\{ x \in \mathbb{R}^d : \left| x^{(j)} - \left(-K - \frac{2}{K} + i \cdot \frac{2}{K} \right) \right| < \delta \right\}. \quad (44)$$

We repeat the whole construction N_n many times. Thus we obtain an approximation $f_{\tilde{w}}$ of

$$N_n \cdot \frac{1}{N_n} \cdot m(x)$$

which satisfies

$$|f_{\tilde{w}}(x) - m(x)| \leq c_{16} \cdot \left(\frac{1}{K} + N_n \cdot (K^2 + 1)^d \cdot \frac{1}{n} \right) \quad (45)$$

for $x \notin A$.

Next we shift the grid along the j -th component so that $[-K, K]^d$ is always covered. This means we modify all $u_i^{(j)}, v_i^{(j)}$ by the same additional summand which is chosen from the set

$$\left\{ k \cdot \frac{2}{K^2} : k = 0, 1, \dots, K - 1 \right\}$$

for fixed $j \in \{1, \dots, d\}$. Thus we obtain K different versions of $f_{\tilde{w}}$ that still satisfy (45) for all $x \in [-K, K]^d$ up to corresponding versions of A .

Since we shift the grid of cubes we obtain for fixed $j \in \{1, \dots, d\}$ K disjoint versions of $\bigcup_{i \in \{0, 1, \dots, K^2 + 1\}} \left\{ x \in \mathbb{R}^d : \left| x^{(j)} - \left(-K - \frac{2}{K} + i \cdot \frac{2}{K} \right) \right| < \delta \right\}$. The sum of $\mathbf{P}_X$ -measures of these K disjoint sets is less than or equal to one. Therefore at least one of them must have measure less than or equal to $\frac{1}{K}$. Consequently we can shift u_i and v_i so that

$$\mathbf{P}_X(A) \leq \sum_{j \in \{1, \dots, d\}} \frac{1}{K} = \frac{d}{K}$$

holds.

Now we have shown that there exists a shifted version of the grid such that the set A has a measure less than or equal to $\frac{d}{K}$. By inequality (45) we obtain that

$$|f_{\tilde{w}}(x) - m(x)| \leq c_{16} \cdot \left(\frac{1}{K} + \frac{N_n \cdot K^{2d}}{n} \right)$$

holds for $x \in [-K, K]^d \setminus A$.

From the second assertion of Lemma 6 in [Drews and Kohler \(2024\)](#) we obtain

$$|f_{\bar{\mathbf{w}}}(x)| \leq \|m\|_{\infty} \cdot \left(3^d + (K^2 + 1)^d \cdot \frac{1}{n}\right)$$

for $x \in \mathbb{R}^d$.

Summarizing the above results we get

$$\begin{aligned} & \int |f_{\bar{\mathbf{w}}}(x) - m(x)|^2 \mathbf{P}_X(dx) \\ &= \int_{[-K, K]^d \setminus A} |f_{\bar{\mathbf{w}}}(x) - m(x)|^2 \mathbf{P}_X(dx) + \int_A |f_{\bar{\mathbf{w}}}(x) - m(x)|^2 \mathbf{P}_X(dx) \\ & \quad + \int_{\mathbb{R}^d \setminus [-K, K]^d} |f_{\bar{\mathbf{w}}}(x) - m(x)|^2 \mathbf{P}_X(dx) \\ &\leq c_{16}^2 \left(\frac{1}{K} + \frac{N_n \cdot K^{2d}}{n} \right)^2 + c_{17} \left(3^d + \frac{K^{2d}}{n} \right)^2 \cdot \frac{d}{K} \\ & \quad + c_{18} \left(3^d + \frac{K^{2d}}{n} \right)^2 \cdot \mathbf{P}_X(\mathbb{R}^d \setminus [-K, K]^d), \end{aligned}$$

which implies the assertion. $\square$

5.2. Proof of Theorem 3.1

Proof. Let $\epsilon > 0$, $K \in \mathbb{N}$ arbitrary and $N_n = \lceil c_{19} \cdot (\log n)^{18L} \rceil$. By Jensen's inequality for conditional expectation we obtain

$$\mathbf{E} \{ |m(X)|^2 \} = \mathbf{E} \{ |\mathbf{E}\{Y|X\}|^2 \} \leq \mathbf{E} \{ \mathbf{E} \{ |Y|^2 | X \} \} = \mathbf{E} \{ Y^2 \} < \infty.$$

Furthermore, let $\bar{m} : \mathbb{R}^d \rightarrow \mathbb{R}$ be a Lipschitz continuous and bounded function such that

$$\int |\bar{m}(x) - m(x)|^2 \mathbf{P}_X(dx) \leq \epsilon. \quad (46)$$

We denote by A_n the event that firstly there exists pairwise disjoint $k_1, \dots, k_{N_n(K^2+1)^d}$ such that the weight vector $\mathbf{w}^{(0)}$ satisfies

$$|(\mathbf{w}^{(0)})_{k_s, i, j}^{(l)} - \mathbf{w}_{s, i, j}^{(l)}| \leq \log n \quad \text{for all } l \in \{0, \dots, L-1\}, s \in \{1, \dots, N_n \cdot (K^2 + 1)^d\}$$

for some weight vector $\mathbf{w}$ which satisfies the conditions (30)–(38) of Theorem 5.5 for $\bar{m}$ and that secondly the inequality

$$\frac{1}{n} \sum_{i=1}^n Y_i^2 \leq \beta_n^3$$

holds.

In case that A_n holds α_k is chosen as in Theorem 5.5 for $\bar{m}$, otherwise we set $\alpha_1 = \dots = \alpha_{N_n(K^2+1)^d} = 0$. Then we define the weight vector $\mathbf{w}^*$ for given $\mathbf{w}$ by

$$(\mathbf{w}^*)_{k, i, j}^{(l)} = \mathbf{w}_{k, i, j}^{(l)} \quad \text{for all } l = 0, \dots, L-1,$$

$$(\mathbf{w}^*)_{1,1,j_k}^{(L)} = \alpha_k \quad \text{for all } k = 1, \dots, N_n \cdot (K^2 + 1)^d,$$

$$(\mathbf{w}^*)_{1,1,k}^{(L)} = 0 \quad \text{for all } k \notin \{j_1, \dots, j_{N_n(K^2+1)^d}\}$$

and $(\mathbf{w}^*)^{(0)}$ by

$$((\mathbf{w}^*)^{(0)})_{k,i,j}^{(l)} = (\mathbf{w}^{(0)})_{k,i,j}^{(l)} \quad \text{for all } l = 0, \dots, L-1,$$

$$((\mathbf{w}^*)^{(0)})_{1,1,j_k}^{(L)} = \alpha_k \quad \text{for all } k = 1, \dots, N_n \cdot (K^2 + 1)^d,$$

$$((\mathbf{w}^*)^{(0)})_{1,1,k}^{(L)} = 0 \quad \text{for all } k \notin \{j_1, \dots, j_{N_n(K^2+1)^d}\}.$$

In the *first step of the proof* we start by decomposing the L_2 error of m_n in a sum of several terms. We have

$$\begin{aligned} & \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) \\ &= (\mathbf{E}\{|m_n(X) - Y|^2 | \mathcal{D}_n\} - \mathbf{E}\{|m(X) - Y|^2\}) \cdot 1_{A_n} \\ & \quad + \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) \cdot 1_{A_n^c} \\ &= \left(\mathbf{E}\{|m_n(X) - Y|^2 | \mathcal{D}_n\} - (1 + \epsilon) \cdot \mathbf{E}\{|m_n(X) - T_{\beta_n} Y|^2 | \mathcal{D}_n\} \right) \cdot 1_{A_n} \\ & \quad + \left((1 + \epsilon) \cdot \mathbf{E}\{|m_n(X) - T_{\beta_n} Y|^2 | \mathcal{D}_n\} - (1 + \epsilon) \cdot \frac{1}{n} \sum_{i=1}^n |m_n(X_i) - T_{\beta_n} Y_i|^2 \right) \cdot 1_{A_n} \\ & \quad + \left((1 + \epsilon) \cdot \frac{1}{n} \sum_{i=1}^n |m_n(X_i) - T_{\beta_n} Y_i|^2 - (1 + \epsilon) \cdot \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}^{(n)}}(X_i) - T_{\beta_n} Y_i|^2 \right) \cdot 1_{A_n} \\ & \quad + \left((1 + \epsilon) \cdot \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}^{(n)}}(X_i) - T_{\beta_n} Y_i|^2 - (1 + \epsilon)^2 \cdot \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}^{(n)}}(X_i) - Y_i|^2 \right) \cdot 1_{A_n} \\ & \quad + \left((1 + \epsilon)^2 \cdot \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}^{(n)}}(X_i) - Y_i|^2 - \mathbf{E}\{|m(X) - Y|^2\} \right) \cdot 1_{A_n} \\ & \quad + \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) \cdot 1_{A_n^c} \\ &= \sum_{j=1}^6 T_{j,n}. \end{aligned}$$

In the *second step of the proof* we show

$$\limsup_{n \rightarrow \infty} \mathbf{E} T_{j,n} \leq 0 \quad \text{for } j \in \{1, 4\}.$$

By using $(a + b)^2 \leq (1 + \epsilon) \cdot a^2 + (1 + \frac{1}{\epsilon}) \cdot b^2$ for $a, b \in \mathbb{R}$ we obtain

$$\mathbf{E} T_{1,n} \leq \left(1 + \frac{1}{\epsilon} \right) \cdot \mathbf{E}\{|T_{\beta_n} Y - Y|^2\}$$

and

$$\mathbf{E}T_{4,n} \leq (1 + \epsilon) \cdot \left(1 + \frac{1}{\epsilon}\right) \cdot \mathbf{E}\{|T_{\beta_n}Y - Y|^2\}.$$

Together with $\beta_n \rightarrow \infty$ ($n \rightarrow \infty$) and $\mathbf{E}Y^2 < \infty$ this implies the assertion of the second step.

In the *third step of the proof* we show

$$\limsup_{n \rightarrow \infty} \mathbf{E}T_{3,n} \leq 0.$$

If $|y| \leq \beta_n$ then it holds

$$|T_{\beta_n}z - y| \leq |z - y|$$

for any $z \in \mathbb{R}$. This implies

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n |m_n(X_i) - T_{\beta_n}Y_i|^2 &= \frac{1}{n} \sum_{i=1}^n |T_{\beta_n}f_{\mathbf{w}^{(t_n)}}(X_i) - T_{\beta_n}Y_i|^2 \\ &\leq \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}^{(t_n)}}(X_i) - T_{\beta_n}Y_i|^2. \end{aligned}$$

Thus the assertion of the third step holds.

In the *fourth step of the proof* we show that the assumptions (15)–(17) of Theorem 5.1 are satisfied which means that

$$\|(\nabla_{\mathbf{w}}F)(\mathbf{w})\| \leq L_n$$

$$\text{for all } \mathbf{w} \in S := \left\{ \mathbf{v} : \|\mathbf{v} - \mathbf{w}^{(0)}\| \leq 2 \cdot \sqrt{F(\mathbf{w}^{(0)})} + 1 \right\},$$

$$\|(\nabla_{\mathbf{w}}F)(\mathbf{w}) - (\nabla_{\mathbf{w}}F)(\tilde{\mathbf{w}})\| \leq L_n \cdot \|\mathbf{w} - \tilde{\mathbf{w}}\|$$

for all $\mathbf{w}, \tilde{\mathbf{w}} \in S$ and

$$\begin{aligned} &|F(\mathbf{w}^*) - F((\mathbf{w}^*)^{(0)})| \\ &\leq D_n \cdot \|((\mathbf{w}^*)^{(L)})_{1,1,k}\|_{k=1,\dots,K_n} \cdot \|(\mathbf{w}_{k,i,j}^{(l)})_{k,i,j,l:l < L} - ((\mathbf{w}^{(0)})^{(l)})_{k,i,j,l:l < L}\| \end{aligned}$$

for all $(\mathbf{w}_{k,i,j}^{(l)})_{k,i,j,l:l < L} \in \tilde{S}$ where

$$\tilde{S} := \left\{ (\tilde{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l:l < L} : \|(\tilde{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l:l < L} - ((\mathbf{w}^{(0)})^{(l)})_{k,i,j,l:l < L}\| \leq \sqrt{2 \cdot F(\mathbf{w}^{(0)})} \right\},$$

provided A_n holds.

If A_n holds, then we also have

$$F_n(\mathbf{w}^{(0)}) = \frac{1}{n} \sum_{i=1}^n Y_i^2 \leq \beta_n^3.$$

Let $\mathbf{w} \in S$. Then we can conclude that

$$\begin{aligned} \|(\mathbf{w}_{k,i,j}^{(l)})_{k,i,j,l:l \leq L}\|_{\infty} &\leq \|\mathbf{w} - \mathbf{w}^{(0)}\| + \|((\mathbf{w}^{(0)})^{(l)})_{k,i,j,l:l \leq L}\|_{\infty} \\ &\leq 2 \cdot \sqrt{F_n(\mathbf{w}^{(0)})} + 1 + c_{20} \cdot (\log n)^2 \end{aligned}$$

$$\leq c_{21} \cdot (\log n)^2$$

and

$$\begin{aligned} \|(\mathbf{w}_{1,1,k}^{(L)})_{k=1,\dots,K_n}\|_\infty &\leq \|\mathbf{w} - \mathbf{w}^{(0)}\| + \|((\mathbf{w}^{(0)})_{1,1,k}^{(L)})_{k=1,\dots,K_n}\|_\infty \\ &\leq 2 \cdot \sqrt{F_n(\mathbf{w}^{(0)})} + 1 \\ &\leq c_{22} \cdot (\log n)^{3/2}. \end{aligned}$$

Hence (21)–(26) are satisfied for $B_n = c_{21} \cdot (\log n)^2$ and $\gamma_n^* = c_{22} \cdot (\log n)^{3/2}$. By Theorem 5.2 and Theorem 5.3 we get for $\alpha_n = c_{23}$ that (15) and (16) are satisfied provided that $L_n \geq K_n^{3/2} \cdot (\log n)^{6L+5}$.

It remains to show that (17) holds. Let $\mathbf{w}$ such that $(\mathbf{w}_{i,j,k}^{(l)})_{i,j,k,l:l < L} \in \tilde{S}$. Then we have

$$\begin{aligned} &|F_n(\mathbf{w}^*) - F_n((\mathbf{w}^*)^{(0)})| \\ &= \left| \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}^*}(X_i) - Y_i|^2 - \frac{1}{n} \sum_{i=1}^n |f_{(\mathbf{w}^*)^{(0)}}(X_i) - Y_i|^2 \right| \\ &= \frac{1}{n} \sum_{i=1}^n \left(f_{\mathbf{w}^*}(X_i) - Y_i + f_{(\mathbf{w}^*)^{(0)}}(X_i) - Y_i \right) \left(f_{\mathbf{w}^*}(X_i) - f_{(\mathbf{w}^*)^{(0)}}(X_i) \right) \\ &\leq \left(\frac{1}{n} \sum_{i=1}^n \left(f_{\mathbf{w}^*}(X_i) - Y_i + f_{(\mathbf{w}^*)^{(0)}}(X_i) - Y_i \right)^2 \right)^{1/2} \\ &\quad \left(\frac{1}{n} \sum_{i=1}^n \left(f_{\mathbf{w}^*}(X_i) - f_{(\mathbf{w}^*)^{(0)}}(X_i) \right)^2 \right)^{1/2} \\ &\leq \left(\frac{2}{n} \sum_{i=1}^n \left(f_{\mathbf{w}^*}(X_i) + f_{(\mathbf{w}^*)^{(0)}}(X_i) \right)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\ &\quad \left(\frac{1}{n} \sum_{i=1}^n \left(\sum_{k=1}^{K_n} \left| (\mathbf{w}^*)_{1,1,k}^{(L)} \right|^2 \cdot \sum_{k=1}^{K_n} \left| f_{\mathbf{w}^*,k,1}^{(L)}(X_i) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(X_i) \right|^2 \right) \right)^{1/2}. \end{aligned}$$

For the first term we get

$$\begin{aligned} &\left(\frac{2}{n} \sum_{i=1}^n \left(f_{\mathbf{w}^*}(X_i) + f_{(\mathbf{w}^*)^{(0)}}(X_i) \right)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\ &= \left(\frac{2}{n} \sum_{i=1}^n \left(f_{(\mathbf{w}^*)^{(0)}}(X_i) - f_{(\mathbf{w}^*)^{(0)}}(X_i) + 2 \cdot f_{(\mathbf{w}^*)^{(0)}}(X_i) \right)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\ &\leq \left(\frac{2}{n} \sum_{i=1}^n 2 \cdot \left(f_{\mathbf{w}^*}(X_i) - f_{(\mathbf{w}^*)^{(0)}}(X_i) \right)^2 + \frac{2}{n} \sum_{i=1}^n 8 \cdot f_{(\mathbf{w}^*)^{(0)}}(X_i)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\ &\leq \left(4 \cdot \frac{1}{n} \sum_{i=1}^n \left(\sum_{k=1}^{K_n} \left| (\mathbf{w}^*)_{1,1,k}^{(L)} \right|^2 \cdot \sum_{k=1}^{K_n} \left| f_{\mathbf{w}^*,k,1}^{(L)}(X_i) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(X_i) \right|^2 \right) \right. \\ &\quad \left. + 16 \cdot \frac{1}{n} \sum_{i=1}^n f_{(\mathbf{w}^*)^{(0)}}(X_i)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \end{aligned}$$

$$\leq \left(4 \cdot \sum_{k=1}^{K_n} \left| (\mathbf{w}^*)_{1,1,k}^{(L)} \right|^2 \cdot \max_{i=1,\dots,n} \sum_{k=1}^{K_n} \left| f_{\mathbf{w}^*,k,1}^{(L)}(X_i) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(X_i) \right|^2 + 16 \cdot \frac{1}{n} \sum_{i=1}^n f_{(\mathbf{w}^*)^{(0)}}(X_i)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2}.$$

By Theorem 5.5 for $\bar{m}$ we get $|\alpha_k| \leq \frac{c_{24}}{N_n}$ and

$$f_{(\mathbf{w}^*)^{(0)}}(X_i) \leq c_{25} \cdot \left(3^d + \frac{(K^2 + 1)^d}{n} \right).$$

From the proof of Theorem 5.2 we know that

$$\begin{aligned} |f_{\mathbf{w}^*,k,1}^{(L)}(x) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(x)| \\ \leq c_{26} \cdot (\log n)^{2L} \cdot \max_{i,j,s:s \leq L} |(\mathbf{w}^*)_{k,i,j}^{(s)} - ((\mathbf{w}^*)^{(0)})_{k,i,j}^{(s)}| \end{aligned} \quad (47)$$

holds. Therefore we obtain for $(\mathbf{w}_{k,i,j}^{(l)})_{k,i,j,l:l \leq L} \in \tilde{S}$

$$\begin{aligned} \max_{i=1,\dots,n} \sum_{k=1}^{K_n} \left| f_{\mathbf{w}^*,k,1}^{(L)}(X_i) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(X_i) \right|^2 \\ \leq c_{27} \cdot (\log n)^{4L} \cdot \sum_{k=1}^{K_n} \max_{i,j,s:s \leq L} |(\mathbf{w}^*)_{k,i,j}^{(s)} - ((\mathbf{w}^*)^{(0)})_{k,i,j}^{(s)}|^2 \\ \leq c_{27} \cdot (\log n)^{4L} \cdot \|(\mathbf{w}_{k,i,j}^{(l)})_{k,i,j,l:l \leq L} - ((\mathbf{w}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:l \leq L}\|^2 \\ \leq c_{27} \cdot (\log n)^{4L+3}. \end{aligned}$$

From this together with the definition of $\mathbf{w}^*$ we can conclude

$$\begin{aligned} & \left(\frac{2}{n} \sum_{i=1}^n \left(f_{\mathbf{w}^*}(X_i) + f_{(\mathbf{w}^*)^{(0)}}(X_i) \right)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\ & \leq \left(4 \cdot \sum_{k=1}^{K_n} \left| (\mathbf{w}^*)_{1,1,k}^{(L)} \right|^2 \cdot c_{27} \cdot (\log n)^{4L+3} + 16 \cdot \frac{1}{n} \sum_{i=1}^n f_{(\mathbf{w}^*)^{(0)}}(X_i)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\ & \leq \left(c_{28} \cdot \frac{N_n \cdot (K^2 + 1)^d}{N_n^2} \cdot (\log n)^{4L+3} + \left(c_{29} \cdot \left(3^d + \frac{(K^2 + 1)^d}{n} \right) \right)^2 + 8 \cdot c_1^3 \cdot (\log n)^3 \right)^{1/2} \\ & \leq c_{30} \left(\frac{(K^2 + 1)^d}{N_n} \cdot (\log n)^{4L+3} + \left(\frac{(K^2 + 1)^d}{n} \right)^2 + (\log n)^3 \right)^{1/2}. \end{aligned}$$

Furthermore, due to (47) we have

$$\begin{aligned} & \left(\frac{1}{n} \sum_{i=1}^n \left(\sum_{k=1}^{K_n} \left| (\mathbf{w}^*)_{1,1,k}^{(L)} \right|^2 \cdot \sum_{k=1}^{K_n} \left| f_{\mathbf{w}^*,k,1}^{(L)}(X_i) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(X_i) \right|^2 \right) \right)^{1/2} \\ & \leq \left(\sum_{k=1}^{K_n} \left| (\mathbf{w}^*)_{1,1,k}^{(L)} \right|^2 \cdot \sum_{i=1,\dots,n} \max_{k=1}^{K_n} \left| f_{\mathbf{w}^*,k,1}^{(L)}(X_i) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(X_i) \right|^2 \right)^{1/2} \end{aligned}$$

$$\leq \|((\mathbf{w}^*)_{1,1,k}^{(L)})_{k=1,\dots,K_n}\| \cdot c_{31} \cdot (\log n)^{2L} \cdot \|(\mathbf{w}_{i,j,k}^{(l)})_{i,j,k,l:l < L} - ((\mathbf{w}^{(0)})_{i,j,k}^{(l)})_{i,j,k,l:l < L}\|.$$

This yields

$$\begin{aligned} & |F_n((\mathbf{w}^*)^{(t)}) - F_n((\mathbf{w}^*)^{(0)})| \\ & \leq c_{32} \left(\frac{(K^2 + 1)^{d/2}}{N_n^{1/2}} \cdot (\log n)^{2L+3/2} + \frac{(K^2 + 1)^d}{n} + (\log n)^{3/2} \right) \\ & \quad (\log n)^{2L} \cdot \|((\mathbf{w}^*)_{1,1,k}^{(L)})_{k=1,\dots,K_n}\| \cdot \|(\mathbf{w}_{i,j,k}^{(l)})_{i,j,k,l:l < L} - ((\mathbf{w}^{(0)})_{i,j,k}^{(l)})_{i,j,k,l:l < L}\|. \end{aligned}$$

Thus (17) is satisfied with

$$D_n = c_{32} \cdot (\log n)^{2L} \cdot \left(\frac{(K^2 + 1)^{d/2}}{N_n^{1/2}} \cdot (\log n)^{2L+3/2} + \frac{(K^2 + 1)^d}{n} + (\log n)^{3/2} \right).$$

For n large we get

$$D_n \leq c_{33} \cdot (\log n)^{4L+2}.$$

In the *fifth step of the proof* we show

$$\mathbf{P}(A_n^c) \leq \frac{c_{34}}{(\log n)^3}.$$

To show this, we first bound the probability that the weight vector $\mathbf{w}^{(0)}$ does not satisfy the first condition of the event A_n . For this, we consider a sequential choice of weights in the K_n fully connected neural networks. Each of these K_n fully connected neural networks contains $(r+1) + (L-2) \cdot r \cdot (r+1) + r \cdot (d+1)$ weights. Therefore, the probability that all these weights satisfy condition (42) for $s = 1$ is bounded from below by

$$\left(\frac{\log n}{40d \cdot (\log n)^2} \right)^{(r+1)+(L-2) \cdot r \cdot (r+1)} \cdot \left(\frac{\log n}{16d \cdot (\log n)^2 \cdot n^\tau} \right)^{r \cdot (d+1)}.$$

Consequently, the probability that this condition is never satisfied in the first $n^{r(d+1)\tau+1}$ many fully connected neural networks for j_1 is for large n bounded from above by

$$\begin{aligned} & \left(1 - \left(\frac{1}{40d \cdot \log n} \right)^{(r+1)+(L-2) \cdot r \cdot (r+1)} \cdot \left(\frac{1}{16d \cdot \log n \cdot n^\tau} \right)^{r \cdot (d+1)} \right)^{n^{r(d+1)\tau+1}} \\ & \leq \left(1 - n^{-r(d+1)\tau-0.5} \right)^{n^{r(d+1)\tau+1}}. \end{aligned}$$

Because of condition (6) we have $K_n \geq N_n \cdot (K^2 + 1)^d \cdot n^{r(d+1)\tau+1}$ for large n . This implies that condition (42) is satisfied for large n outside of an event of probability

$$\begin{aligned} & N_n \cdot (K^2 + 1)^d \cdot \left(1 - n^{-r(d+1)\tau-0.5} \right)^{n^{r(d+1)\tau+1}} \\ & \leq N_n \cdot (K^2 + 1)^d \cdot \left(\exp \left(-n^{-r(d+1)\tau-0.5} \right) \right)^{n^{r(d+1)\tau+1}} \\ & \leq N_n \cdot (K^2 + 1)^d \cdot \exp \left(-n^{0.5} \right) \\ & \leq n^\kappa \cdot \exp \left(-n^{0.5} \right) \end{aligned}$$

$$\leq \frac{c_{35}}{n}.$$

Then we obtain for large n by Markov's inequality

$$\begin{aligned} \mathbf{P}(A^c) &\leq \frac{c_{35}}{n} + \mathbf{P}\left\{\frac{1}{n} \sum_{i=1}^n Y_i^2 > \beta_n^3\right\} \\ &\leq \frac{c_{35}}{n} + \frac{\mathbf{E}\{\frac{1}{n} \sum_{i=1}^n Y_i^2\}}{\beta_n^3} \\ &\leq \frac{c_{35}}{n} + \frac{\mathbf{E}\{Y^2\}}{\beta_n^3} \\ &\leq \frac{c_{36}}{(\log n)^3} \end{aligned}$$

where the last inequality holds since $\mathbf{E}Y^2 < \infty$.

In the *sixth step of the proof* we show

$$\limsup_{n \rightarrow \infty} \mathbf{E}T_{2,n} \leq 0.$$

We have

$$\begin{aligned} \frac{1}{1+\epsilon} \cdot \mathbf{E}\{T_{2,n}\} &\leq \int_0^{4 \cdot \beta_n^2} \mathbf{P}\left\{\left(\mathbf{E}\{|m_n(X) - T_{\beta_n} Y|^2 | \mathcal{D}_n\}\right. \right. \\ &\quad \left. \left. - \frac{1}{n} \sum_{i=1}^n |m_n(X_i) - T_{\beta_n} Y_i|^2\right) \cdot 1_{A_n} > t\right\} dt \\ &\leq n^{\frac{-1}{4(d+2)}} + \int_{n^{\frac{-1}{4(d+2)}}}^{4 \cdot \beta_n^2} \mathbf{P}\left\{\left(\mathbf{E}\{|m_n(X) - T_{\beta_n} Y|^2 | \mathcal{D}_n\}\right. \right. \\ &\quad \left. \left. - \frac{1}{n} \sum_{i=1}^n |m_n(X_i) - T_{\beta_n} Y_i|^2\right) \cdot 1_{A_n} > t\right\} dt. \end{aligned}$$

W.l.o.g. we can assume that A_n holds. Hence, by Theorem 5.1, it follows that

$$\|((\mathbf{w}^{(t_n)})_{k,i,j}^{(l)})_{k,i,j,l:l < L} - ((\mathbf{w}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:l < L}\| \leq c_{37} \cdot (\log n)^{3/2}$$

and

$$\|((\mathbf{w}^{(t_n)})_{1,1,k}^{(L)})_{k=1,\dots,K_n}\| \leq c_{38} \cdot (\log n)^{3/2}.$$

hold. Consequently, we obtain

$$\begin{aligned} &\|((\mathbf{w}^{(t_n)})_{k,i,j}^{(l)})_{k,i,j,l:1 \leq l < L}\|_{\infty} \\ &\leq \|((\mathbf{w}^{(t_n)})_{k,i,j}^{(l)})_{k,i,j,l:1 \leq l < L} - ((\mathbf{w}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:1 \leq l < L}\|_{\infty} + \|((\mathbf{w}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:1 \leq l < L}\|_{\infty} \\ &\leq c_{37} \cdot (\log n)^{3/2} + c_{39} \cdot (\log n)^2 \\ &\leq c_{40} \cdot (\log n)^2 \end{aligned}$$

and

$$\|((\mathbf{w}^{(t_n)})_{k,i,k}^{(0)})_{k,i,j}\|_{\infty}$$

$$\begin{aligned}
&\leq \|((\mathbf{w}^{(t_n)})_{k,i,j}^{(0)})_{k,i,j} - ((\mathbf{w}^{(0)})_{k,i,j}^{(0)})_{k,i,j}\|_\infty + \|((\mathbf{w}^{(0)})_{k,i,j}^{(0)})_{k,i,j}\|_\infty \\
&\leq c_{37} \cdot (\log n)^{3/2} + c_{41} \cdot (\log n)^2 \cdot n^\tau \\
&\leq c_{42} \cdot (\log n)^2 \cdot n^\tau.
\end{aligned}$$

This implies that m_n is contained in the function space

$$\{T_{\beta_n} f : f \in \mathcal{F}\}$$

where $\mathcal{F}$ is defined as in Lemma 5.4 with $C = c_{38} \cdot K_n \cdot (\log n)^{3/2}$, $B = c_{40} \cdot (\log n)^2$ and $A = c_{42} \cdot (\log n)^2 \cdot n^\tau$. Thus, with Lemma 5.4 and standard bounds of empirical process theory (cf., Theorem 9.1 in Györfi et al. (2002)), it follows

$$\begin{aligned}
&\mathbf{P}\left\{(\mathbf{E}\{|m_n(X) - T_{\beta_n} Y|^2(X)|\mathcal{D}_n\}) - \frac{1}{n} \sum_{i=1}^n |m_n(X_i) - T_{\beta_n} Y_i|^2 \cdot 1_{A_n} > t\right\} \\
&\leq 8 \cdot \left(c_{43} \cdot \frac{\beta_n}{t/8}\right)^{c_{44} \cdot (\log n)^{c_{45}} \cdot n^{\tau \cdot d} \cdot \left(\frac{c_{46} \cdot K_n \cdot (\log n)^{3/2}}{t/8}\right)^{d/k} + c_{47}} \cdot \exp\left(-\frac{n \cdot t^2}{128 \cdot \beta_n^4}\right).
\end{aligned}$$

Using (5) and $\tau = \frac{1}{d+1}$ we get for $k > (\kappa + 1) \cdot d \cdot (d + 1) \cdot (d + 2)$ and $t > n^{\frac{-1}{4(d+2)}}$ that the left hand side above is for n large enough less than or equal to

$$\begin{aligned}
&8 \cdot \left(c_{43} \cdot \frac{\beta_n}{t/8}\right)^{c_{44} \cdot (\log n)^{c_{45}} \cdot n^{\tau \cdot d} \cdot \left(\frac{c_{46} \cdot K_n \cdot (\log n)^{3/2}}{t/8}\right)^{d/k} + c_{47}} \\
&\quad \cdot \exp\left(-\frac{n \cdot t^2}{256 \cdot \beta_n^4}\right) \cdot \exp\left(-\frac{n \cdot t^2}{256 \cdot \beta_n^4}\right) \\
&\leq \exp\left(c_{48} \cdot (\log n)^{c_{45}} \cdot n^{\tau \cdot d + (\kappa+1) \cdot \frac{d}{k}} \cdot \log\left(c_{43} \cdot \frac{\beta_n}{t/8}\right)\right) \\
&\quad \cdot \exp\left(-\frac{n \cdot t^2}{256 \cdot \beta_n^4}\right) \cdot \exp\left(-\frac{n \cdot t^2}{256 \cdot \beta_n^4}\right) \\
&\leq \exp\left(c_{49} \cdot \left((\log n)^{c_{50}} \cdot n^{\frac{d}{d+1} + (\kappa+1) \cdot \frac{d}{k}} - \frac{n^{\frac{2d+3}{2(d+2)}}}{(\log n)^4}\right)\right) \cdot \exp\left(-\frac{n \cdot t^2}{256 \cdot \beta_n^4}\right) \\
&\leq \exp\left(c_{49} \cdot \left((\log n)^{c_{50}} \cdot n^{\frac{d+1}{d+2}} - \frac{n^{\frac{2d+3}{2(d+2)}}}{(\log n)^4}\right)\right) \cdot \exp\left(-\frac{n \cdot t^2}{256 \cdot \beta_n^4}\right) \\
&= \exp\left(c_{49} \cdot n^{\frac{d+1}{d+2}} \cdot \left((\log n)^{c_{50}} - \frac{n^{\frac{1}{2(d+2)}}}{(\log n)^4}\right)\right) \cdot \exp\left(-\frac{n \cdot t^2}{256 \cdot \beta_n^4}\right) \\
&\leq c_{51} \cdot \exp\left(-\frac{n \cdot t^2}{256 \cdot \beta_n^4}\right) \\
&\leq c_{52} \cdot \exp\left(-\frac{n^{\frac{2d+3}{2(d+2)}}}{256 \cdot \beta_n^4}\right)
\end{aligned}$$

holds. Therefore, we obtain

$$\mathbf{E}\{T_{2,n}\} \leq (1 + \epsilon) \cdot \left(n^{\frac{-1}{4(d+2)}} + c_{53} \cdot \beta_n^2 \cdot \exp\left(-\frac{n^{\frac{2d+3}{2(d+2)}}}{256 \cdot \beta_n^4}\right)\right) \rightarrow 0 \quad (n \rightarrow \infty).$$

In the *seventh step of the proof* we show

$$\limsup_{n \rightarrow \infty} \mathbf{E}\{T_{6,n}\} \leq 0.$$

Due to the assertion of the fifth step together with the integrability of m we get

$$\begin{aligned} \mathbf{E}\{T_{6,n}\} &\leq \left(2 \cdot \int |m_n(x)|^2 \mathbf{P}_X(dx) + 2 \cdot \int |m(x)|^2 \mathbf{P}_X(dx) \right) \cdot \mathbf{P}(A_n^c) \\ &\leq 2 \cdot (\beta_n^2 + c_{54}) \cdot \frac{c_{36}}{(\log n)^3}. \end{aligned}$$

This implies the assertion of the seventh step.

In the *eighth step of the proof* we bound

$$\mathbf{E}T_{5,n}.$$

If A_n holds, then as shown in step four, we can apply Lemma 5.1 with

$$D_n \leq c_{33} \cdot (\log n)^{4L+2}.$$

This together with the definition of $\mathbf{w}^*$ and $(\mathbf{w}^*)^{(0)}$ yields

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n |f_{\mathbf{w}^{(t_n)}}(X_i) - Y_i|^2 &= F_n(\mathbf{w}^{(t_n)}) \\ &\leq F_n((\mathbf{w}^*)^{(0)}) + D_n \cdot \|((\mathbf{w}^*)_{1,1,k}^{(L)})_{k=1,\dots,K_n}\| \cdot \sqrt{2 \cdot F_n(\mathbf{w}^{(0)})} \\ &\quad + \frac{\|((\mathbf{w}^*)_{1,1,k}^{(L)})_{k=1,\dots,K_n} - ((\mathbf{w}^{(0)})_{1,1,k}^{(L)})_{k=1,\dots,K_n}\|^2}{2} + \frac{F_n(\mathbf{w}^{(0)})}{t_n} \\ &\leq F_n((\mathbf{w}^*)^{(0)}) + c_{33} \cdot (\log n)^{4L+2} \cdot \frac{N_n^{1/2} \cdot (K^2 + 1)^{d/2}}{N_n} \cdot (\log n)^{3/2} \\ &\quad + \frac{N_n \cdot (K^2 + 1)^d}{2 \cdot N_n^2} + \frac{c_1^3 \cdot (\log n)^3}{t_n} \\ &\leq F_n((\mathbf{w}^*)^{(0)}) + c_{55} \cdot \frac{(K^2 + 1)^d}{N_n^{1/2}} \cdot (\log n)^{4L+4}. \end{aligned}$$

Thus we obtain

$$\begin{aligned} \mathbf{E}\{T_{5,n}\} &= (1 + \epsilon)^2 \cdot \mathbf{E}\left\{ \left(F_n(\mathbf{w}^{(t_n)}) - \mathbf{E}\{|m(X) - Y|^2\} \right) \cdot 1_{A_n} \right\} \\ &\quad + ((1 + \epsilon)^2 - 1) \cdot \mathbf{E}\left\{ \mathbf{E}\{|m(X) - Y|^2\} \cdot 1_{A_n} \right\} \\ &\leq (1 + \epsilon)^2 \cdot \left(\mathbf{E}\left\{ F_n((\mathbf{w}^*)^{(0)}) \cdot 1_{A_n} + c_{55} \cdot \left(\frac{(K^2 + 1)^d}{N_n^{1/2}} \cdot (\log n)^{4L+4} \right) \cdot 1_{A_n} \right\} \right. \\ &\quad \left. - \mathbf{E}\{|m(X) - Y|^2\} \cdot \mathbf{P}(A_n) \right) + ((1 + \epsilon)^2 - 1) \cdot \mathbf{E}\{|m(X) - Y|^2\}. \end{aligned}$$

Due to the definition of N_n we obtain

$$\frac{(K^2 + 1)^d}{N_n^{1/2}} \cdot (\log n)^{4L+4} \rightarrow 0 \quad (n \rightarrow \infty).$$

Let $\tilde{A}_n$ be the event where the weight vector $\mathbf{w}^{(0)}$ satisfies

$$|(\mathbf{w}^{(0)})_{k_s, i, j}^{(l)} - \mathbf{w}_{s, i, j}^{(l)}| \leq \log n \quad \text{for all } l \in \{0, \dots, L-1\}, s \in \{1, \dots, N_n \cdot (K^2 + 1)^d\}$$

for some weight vector $\mathbf{w}$ which satisfies conditions (30)–(38) of Lemma 5.5 for $\bar{m}$. Then we get from the fifth step of the proof

$$\mathbf{P}(\tilde{A}_n) - \mathbf{P}(A_n) \leq \mathbf{P}\left\{\frac{1}{n} \sum_{i=1}^n Y_i^2 > \beta_n^3\right\} \leq \frac{c_{56}}{(\log n)^3}.$$

This together with the fact that $(X_1, Y_1), \dots, (X_n, Y_n)$ are independent of $\tilde{A}_n$ yields

$$\begin{aligned} & \mathbf{E}\{F_n((\mathbf{w}^*)^{(0)}) \cdot 1_{A_n}\} - \mathbf{E}\{|m(X) - Y|^2\} \cdot \mathbf{P}(A_n) \\ & \leq \mathbf{E}\left\{\frac{1}{n} \sum_{i=1}^n |f_{((\mathbf{w}^*)^{(0)})}(X_i) - Y_i|^2 \cdot 1_{\tilde{A}_n}\right\} - \mathbf{E}\{|m(X) - Y|^2\} \cdot \mathbf{P}(\tilde{A}_n) \\ & \quad + \mathbf{E}\{|m(X) - Y|^2\} \cdot (\mathbf{P}(\tilde{A}_n) - \mathbf{P}(A_n)) \\ & \leq \mathbf{E}\left\{\mathbf{E}\left\{\frac{1}{n} \sum_{i=1}^n |f_{((\mathbf{w}^*)^{(0)})}(X_i) - Y_i|^2 \cdot 1_{\tilde{A}_n} \mid (\mathbf{w}^*)^{(0)}\right\} - \mathbf{E}\{|m(X) - Y|^2\} \cdot 1_{\tilde{A}_n}\right\} \\ & \quad + \frac{c_{57}}{(\log n)^3} \\ & = \mathbf{E}\left\{\left(\mathbf{E}\left\{\frac{1}{n} \sum_{i=1}^n |f_{((\mathbf{w}^*)^{(0)})}(X_i) - Y_i|^2 \mid (\mathbf{w}^*)^{(0)}\right\} - \mathbf{E}\{|m(X) - Y|^2\}\right) \cdot 1_{\tilde{A}_n}\right\} \\ & \quad + \frac{c_{57}}{(\log n)^3} \\ & \leq \mathbf{E}\left\{\int |f_{((\mathbf{w}^*)^{(0)})}(x) - m(x)|^2 \mathbf{P}_X(dx) \cdot 1_{\tilde{A}_n}\right\} + \frac{c_{57}}{(\log n)^3}. \end{aligned}$$

Because of the choice of $\bar{m}$ and Theorem 5.5 we get for K such large that $\text{supp}(X) \subseteq [-K, K]^d$

$$\begin{aligned} & \mathbf{E}\left\{\int |f_{((\mathbf{w}^*)^{(0)})}(x) - m(x)|^2 \mathbf{P}_X(dx) \cdot 1_{\tilde{A}_n}\right\} \\ & \leq 2 \cdot \mathbf{E}\left\{\int |f_{((\mathbf{w}^*)^{(0)})}(x) - \bar{m}(x)|^2 \mathbf{P}_X(dx) \cdot 1_{\tilde{A}_n}\right\} + 2 \int |\bar{m}(x) - m(x)|^2 \mathbf{P}_X(dx) \\ & \leq c_{58} \cdot \left(\frac{1}{K} + \frac{N_n^2 \cdot K^{4d}}{n^2} + \left(\frac{K^{2d}}{n} + 1\right)^2 \cdot \mathbf{P}_X(\mathbb{R}^d \setminus [-K, K]^d)\right) + 2\epsilon \\ & \leq c_{59} \cdot \left(\frac{1}{K} + \frac{N_n^2 \cdot K^{4d}}{n^2}\right) + 2\epsilon. \end{aligned}$$

Summarizing the above results yields

$$\limsup_{n \rightarrow \infty} \mathbf{E}\{T_{5,n}\} \leq c_{60} \cdot (1 + \epsilon)^2 \cdot \left(\frac{1}{K} + 2\epsilon\right) + ((1 + \epsilon)^2 - 1) \cdot \mathbf{E}\{|m(X) - Y|^2\}.$$

In the *ninth step of the proof* we finish the proof of Theorem 3.1. The results of steps 1,2,3,6,7 and 8 imply for $K \rightarrow \infty$

$$\limsup_{n \rightarrow \infty} \mathbf{E} \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx)$$

$$\leq c_{61} \cdot \left((1 + \epsilon)^2 \cdot 2\epsilon + ((1 + \epsilon)^2 - 1) \cdot \mathbf{E}\{|m(X) - Y|^2\} \right).$$

With $\epsilon \rightarrow 0$ we get the assertion. $\square$

5.3. Auxiliary results for the proof of Theorem 3.2

The following theorem is crucial for proving Theorem 3.2 and Theorem 3.3. It is needed to analyze the rate of convergence of the over-parametrized deep neural network estimate.

Theorem 5.6. *Let $n \in \mathbb{N}$ with $n \geq 2$, let $(X, Y), (X_1, Y_1), \dots, (X_n, Y_n)$ be independent and identically distributed $\mathbb{R}^d \times \mathbb{R}$ -valued random variables such that $\text{supp}(X)$ is bounded, that*

$$\mathbf{E} \left\{ e^{c_3 \cdot Y^2} \right\} < \infty \quad (48)$$

holds and that the corresponding regression function $m(x) = \mathbf{E}\{Y|X = x\}$ is bounded. Let $\sigma(x) = 1/(1 + e^{-x})$ be the logistic squasher, let $K_n, L, r, t_n \in \mathbb{N}$, $M_n \geq 1$ and $\lambda_n, \tau > 0$. Let $\tilde{K}_n \in \{1, \dots, K_n\}$,

$$w_{k,i,j}^{(l)} \in [-20d \cdot (\log n)^2, 20d \cdot (\log n)^2] \quad \text{for } l = 1, \dots, L, \quad k = 1, \dots, \tilde{K}_n$$

and

$$w_{k,i,j}^{(0)} \in [-8d \cdot (\log n)^2 \cdot n^\tau, 8d \cdot (\log n)^2 \cdot n^\tau] \quad \text{for } k = 1, \dots, \tilde{K}_n.$$

Assume that

$$\sqrt{\sum_{k=1}^{\tilde{K}_n} |w_{1,1,k}^{(L)}|^2} \leq M_n, \quad (49)$$

and that

$$\left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} \cdot f_{\bar{\mathbf{w}},k,1}^{(L)}(x) \right| \leq \beta_n \quad (50)$$

holds for $x \in \text{supp}(X)$ and for all $\bar{\mathbf{w}}$ which satisfy

$$|\bar{w}_{k,i,j}^{(l)} - w_{k,i,j}^{(l)}| \leq \log n \quad \text{for } l = 0, \dots, L-1.$$

Assume further

$$\frac{K_n}{n^\kappa} \rightarrow 0 \quad (n \rightarrow \infty) \quad (51)$$

for some $\kappa > 0$, and

$$\frac{K_n}{\tilde{K}_n \cdot n^{r \cdot (d+1) \cdot \tau + 1}} \rightarrow \infty \quad (n \rightarrow \infty) \quad (52)$$

Define the estimate m_n as in Section 2 with

$$\lambda_n = \frac{1}{t_n} \quad \text{and} \quad t_n = \lceil c_{62} \cdot L_n \rceil \quad (53)$$

for some $c_{62} \geq 1$ and for some $L_n > 0$ which satisfies

$$L_n \geq K_n^{3/2} \cdot (\log n)^{6L+2},$$

and assume

$$c_1 \cdot c_3 \geq 2. \quad (54)$$

Then we have for any $\epsilon > 0$

$$\begin{aligned} \mathbf{E} \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) &\leq c_{63} \cdot \left(\frac{n^{\tau \cdot d + \epsilon}}{n} + M_n^2 \cdot (\log n)^{4L+3/2} \right. \\ &\quad + \sup_{\substack{(\bar{w}_{k,i,j}^{(l)})_{k,i,j,l} \\ |\bar{w}_{k,i,j}^{(l)} - w_{k,i,j}^{(l)}| \leq \log n \ (l=0, \dots, L-1)}} \int \left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} \cdot f_{\tilde{w},k,1}^{(L)}(x) - m(x) \right|^2 \mathbf{P}_X(dx) \Bigg). \end{aligned}$$

Proof. Let A_n be the event that firstly the weight vector $\mathbf{w}^{(0)}$ satisfies

$$|(\mathbf{w}^{(0)})_{k_s,i,j}^{(l)} - \mathbf{w}_{s,i,j}^{(l)}| \leq \log n \quad \text{for all } l \in \{0, \dots, L-1\}, s \in \{1, \dots, \tilde{K}_n\}$$

for some pairwise distinct $k_1, \dots, k_{\tilde{K}_n} \in \{1, \dots, K_n\}$ and that secondly

$$\max_{i=1, \dots, n} |Y_i| \leq \sqrt{\beta_n}$$

holds. Then we define the weight vectors $\mathbf{w}^*$ for given $\tilde{\mathbf{w}}$ by

$$(\mathbf{w}^*)_{k,i,j}^{(l)} = \tilde{\mathbf{w}}_{k,i,j}^{(l)} \quad \text{for all } l = 0, \dots, L-1,$$

$$(\mathbf{w}^*)_{1,1,j_k}^{(L)} = \tilde{\mathbf{w}}_{1,1,k}^{(L)} \quad \text{for all } k = 1, \dots, \tilde{K}_n,$$

$$(\mathbf{w}^*)_{1,1,k}^{(L)} = 0 \quad \text{for all } k \notin \{j_1, \dots, j_{\tilde{K}_n}\}$$

and $(\mathbf{w}^*)^{(0)}$ by

$$((\mathbf{w}^*)^{(0)})_{k,i,j}^{(l)} = (\mathbf{w}_{k,i,j}^{(0)})^{(l)} \quad \text{for all } l = 0, \dots, L-1,$$

$$(\mathbf{w}^*)_{1,1,j_k}^{(0)(L)} = (\mathbf{w}_{1,1,k}^{(0)})^{(L)} \quad \text{for all } k = 1, \dots, \tilde{K}_n,$$

$$((\mathbf{w}^*)^{(0)})_{1,1,k}^{(L)} = 0 \quad \text{for all } k \notin \{j_1, \dots, j_{\tilde{K}_n}\}.$$

In the following we set

$$m_{\beta_n}(x) = \mathbf{E}\{T_{\beta_n} Y | X = x\}.$$

Furthermore, we assume w.l.o.g. that n is sufficiently large and that $\|m\|_\infty \leq \beta_n$ holds.

In the *first step of the proof* we decompose the L_2 error of m_n in a sum of several terms. We have

$$\int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx)$$

$$\begin{aligned}
&= \left(\mathbf{E} \{ |m_n(X) - Y|^2 | \mathcal{D}_n \} - \mathbf{E} \{ |m(X) - Y|^2 \} \right) \cdot 1_{A_n} \\
&\quad + \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) \cdot 1_{A_n^c} \\
&= \left[\mathbf{E} \{ |m_n(X) - Y|^2 | \mathcal{D}_n \} - \mathbf{E} \{ |m(X) - Y|^2 \} \right. \\
&\quad \left. - \left(\mathbf{E} \{ |m_n(X) - T_{\beta_n} Y|^2 | \mathcal{D}_n \} - \mathbf{E} \{ |m_{\beta_n}(X) - T_{\beta_n} Y|^2 \} \right) \right] \cdot 1_{A_n} \\
&\quad + \left[\mathbf{E} \{ |m_n(X) - T_{\beta_n} Y|^2 | \mathcal{D}_n \} - \mathbf{E} \{ |m_{\beta_n}(X) - T_{\beta_n} Y|^2 \} \right. \\
&\quad \left. - 2 \cdot \frac{1}{n} \sum_{i=1}^n \left(|m_n(X_i) - T_{\beta_n} Y_i|^2 - |m_{\beta_n}(X_i) - T_{\beta_n} Y_i|^2 \right) \right] \cdot 1_{A_n} \\
&\quad + \left[2 \cdot \frac{1}{n} \sum_{i=1}^n |m_n(X_i) - T_{\beta_n} Y_i|^2 - 2 \cdot \frac{1}{n} \sum_{i=1}^n |m_{\beta_n}(X_i) - T_{\beta_n} Y_i|^2 \right. \\
&\quad \left. - \left(2 \cdot \frac{1}{n} \sum_{i=1}^n |m_n(X_i) - Y_i|^2 - 2 \cdot \frac{1}{n} \sum_{i=1}^n |m(X_i) - Y_i|^2 \right) \right] \cdot 1_{A_n} \\
&\quad + \left[2 \cdot \frac{1}{n} \sum_{i=1}^n |m_n(X_i) - Y_i|^2 - 2 \cdot \frac{1}{n} \sum_{i=1}^n |m(X_i) - Y_i|^2 \right] \cdot 1_{A_n} \\
&\quad + \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) \cdot 1_{A_n^c} \\
&=: \sum_{j=1}^5 T_{j,n}.
\end{aligned}$$

In the *second step of the proof* we show

$$\mathbf{E}T_{1,n} \leq c_{64} \cdot \frac{\log n}{n} \quad \text{and} \quad \mathbf{E}T_{3,n} \leq c_{65} \cdot \frac{\log n}{n}.$$

This follows as in the proof of Lemma 1 in [Bauer and Kohler \(2019\)](#).

In the *third step of the proof* we show

$$\mathbf{E}T_{5,n} \leq c_{66} \cdot \frac{(\log n)^2}{n}.$$

Due to the definition of m_n and the assumption that $\|m\|_\infty \leq \beta_n$, it holds

$$\int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) \leq 4 \cdot c_1^2 \cdot (\log n)^2.$$

Thus, it suffices to show

$$\mathbf{P}(A_n^c) \leq \frac{c_{67}}{n}. \quad (55)$$

To do this, we first bound the probability that the weights in the first of the K_n fully connected neural networks differ by at most $\log n$ from $(\mathbf{w}_{1,i,j}^{(l)})_{i,j,l:l < L}$ in all components. For large n , this probability is bounded from below by

$$\left(\frac{\log n}{40d \cdot (\log n)^2} \right)^{r \cdot (r+1) \cdot (L-2)} \cdot \left(\frac{\log n}{16d \cdot (\log n)^2 \cdot n^\tau} \right)^{r \cdot (d+1)} \geq n^{-r \cdot (d+1) \cdot \tau - 0.5}.$$

Then the probability that none of the first $n^{r \cdot (d+1) \cdot \tau + 1}$ neural networks satisfies this condition is bounded above by

$$\begin{aligned} (1 - n^{-r \cdot (d+1) \cdot \tau - 0.5})^{n^{r \cdot (d+1) \cdot \tau + 1}} &\leq \left(\exp \left(-n^{-r \cdot (d+1) \cdot \tau - 0.5} \right) \right)^{n^{r \cdot (d+1) \cdot \tau + 1}} \\ &= \exp(-n^{0.5}). \end{aligned}$$

Assumption (52) implies $K_n \geq n^{r \cdot (d+1) \cdot \tau + 1} \cdot \tilde{K}_n$ for large n . Thus, we can apply this construction successively for all $\tilde{K}_n$ weights $((\mathbf{w}^{(0)})_{k,i,j}^{(l)})_{i,j,l:l < L}$. The probability that there exists $k \in \{1, \dots, \tilde{K}_n\}$ such that none of the K_n weight vectors of the fully connected neural network differs from $(\mathbf{w}_{k,i,j}^{(l)})_{i,j,l:l < L}$ by at most $\log n$ is then for large n bounded from above by

$$\tilde{K}_n \cdot \exp(-n^{0.5}) \leq n^\kappa \cdot \exp(-n^{0.5}) \leq \frac{c_{68}}{n}.$$

Hence, for large n it is

$$\begin{aligned} \mathbf{P}(A_n^c) &\leq \frac{c_{68}}{n} + \mathbf{P}\left\{ \max_{i=1, \dots, n} |Y_i| > \sqrt{\beta_n} \right\} \leq \frac{c_{68}}{n} + n \cdot \mathbf{P}\{|Y| > \sqrt{\beta_n}\} \\ &\leq \frac{c_{68}}{n} + n \cdot \frac{\mathbf{E}\{\exp(c_3 \cdot Y^2)\}}{\exp(c_3 \cdot \beta_n)} \leq \frac{c_{69}}{n}, \end{aligned}$$

where the last inequality holds because of (48) and (54).

In the *fourth step of the proof* we show that the assumptions (15)–(17) of Lemma 5.1 are satisfied which means that

$$\|(\nabla_{\mathbf{w}} F)(\mathbf{w})\| \leq L_n$$

for all $\mathbf{w} \in S := \left\{ \mathbf{v} : \|\mathbf{v} - \mathbf{w}^{(0)}\| \leq 2 \cdot \sqrt{F(\mathbf{w}^{(0)})} + 1 \right\}$,

$$\|(\nabla_{\mathbf{w}} F)(\mathbf{w}) - (\nabla_{\mathbf{w}} F)(\bar{\mathbf{w}})\| \leq L_n \cdot \|\mathbf{w} - \bar{\mathbf{w}}\|$$

for all $\mathbf{w}, \bar{\mathbf{w}} \in S$ and

$$\begin{aligned} &|F(\mathbf{w}^*) - F((\mathbf{w}^*)^{(0)})| \\ &\leq D_n \cdot \|((\mathbf{w}^*)_{1,1,k}^{(L)})_{k=1, \dots, K_n}\| \cdot \|(\mathbf{w}_{i,j,k}^{(l)})_{i,j,k,l:l < L} - ((\mathbf{w}^{(0)})_{i,j,k}^{(l)})_{i,j,k,l:l < L}\| \end{aligned}$$

for all

$$\begin{aligned} &(\mathbf{w}_{k,i,j}^{(l)})_{k,i,j,l:l < L} \in \tilde{S} \\ &:= \left\{ (\bar{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l:l < L} : \|(\bar{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l:l < L} - ((\mathbf{w}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:l < L}\| \leq \sqrt{2 \cdot F(\mathbf{w}^{(0)})} \right\} \end{aligned}$$

hold, provided A_n holds.

If A_n holds, then we have

$$F_n(\mathbf{w}^{(0)}) = \frac{1}{n} \sum_{i=1}^n Y_i^2 \leq \beta_n.$$

Let $\mathbf{w} \in S$. Then we have

$$\|(\mathbf{w}_{k,i,j}^{(l)})_{k,i,j,l:l < L}\|_\infty \leq \|\mathbf{w} - \mathbf{w}^{(0)}\| + \|((\mathbf{w}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:l < L}\|_\infty$$

$$\begin{aligned}
&\leq 2 \cdot \sqrt{F_n(\mathbf{w}^{(0)})} + 1 + c_{70} \cdot (\log n)^2 \\
&\leq c_{71} \cdot (\log n)^2
\end{aligned}$$

and

$$\begin{aligned}
\|(\mathbf{w}_{1,1,k}^{(L)})_{k=1,\dots,K_n}\|_\infty &\leq \|\mathbf{w} - \mathbf{w}^{(0)}\| + \|((\mathbf{w}^{(0)})^{(L)})_{1,1,k})_{k=1,\dots,K_n}\|_\infty \\
&\leq 2 \cdot \sqrt{F_n(\mathbf{w}^{(0)})} + 1 \\
&\leq c_{72} \cdot (\log n)^{1/2}.
\end{aligned}$$

Hence (21)–(26) are satisfied for $B_n = c_{71} \cdot (\log n)^2$ and $\gamma_n^* = c_{72} \cdot (\log n)^{1/2}$. By Lemma 5.2 and Lemma 5.3 we get that (15) and (16) are satisfied provided that $L_n \geq K_n^{3/2} \cdot (\log n)^{6L+2}$.

Furthermore, let $\tilde{\mathbf{w}}$ such that $(\tilde{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l:l < L} \in \tilde{\mathcal{S}}$. Then, as in the proof of Theorem 3.1, we obtain

$$\begin{aligned}
&|F_n(\mathbf{w}^*) - F_n((\mathbf{w}^*)^{(0)})| \\
&\leq \left(\frac{2}{n} \sum_{i=1}^n \left(f_{\mathbf{w}^*}(X_i) + f_{(\mathbf{w}^*)^{(0)}}(X_i) \right)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\
&\quad \left(\frac{1}{n} \sum_{i=1}^n \left(\sum_{k=1}^{K_n} |(\mathbf{w}^*)_{1,1,k}|^2 \cdot \sum_{k=1}^{K_n} \left| f_{\mathbf{w}^*,k,1}^{(L)}(X_i) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(X_i) \right|^2 \right) \right)^{1/2}.
\end{aligned}$$

Considering that

$$\begin{aligned}
&\left(\frac{1}{n} \sum_{i=1}^n \left(\sum_{k=1}^{K_n} |(\mathbf{w}^*)_{1,1,k}|^2 \cdot \sum_{k=1}^{K_n} \left| f_{\mathbf{w}^*,k,1}^{(L)}(X_i) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(X_i) \right|^2 \right) \right)^{1/2} \\
&\leq \|((\mathbf{w}^*)^{(L)})_{1,1,k})_{k=1,\dots,K_n}\| \cdot c_{73} \cdot (\log n)^{2L} \cdot \|(\tilde{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l:l < L} - ((\tilde{\mathbf{w}}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:l < L}\|
\end{aligned}$$

and since $(\tilde{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l:l < L} \in \tilde{\mathcal{S}}$ we get

$$\begin{aligned}
&\left(\frac{2}{n} \sum_{i=1}^n \left(f_{\mathbf{w}^*}(X_i) + f_{(\mathbf{w}^*)^{(0)}}(X_i) \right)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\
&\leq \left(4 \cdot \sum_{k=1}^{K_n} |(\mathbf{w}^*)_{1,1,k}^{(L)}|^2 \cdot \max_{i=1,\dots,n} \sum_{k=1}^{K_n} \left| f_{\mathbf{w}^*,k,1}^{(L)}(X_i) - f_{(\mathbf{w}^*)^{(0)},k,1}^{(L)}(X_i) \right|^2 \right. \\
&\quad \left. + 16 \cdot \frac{1}{n} \sum_{i=1}^n f_{(\mathbf{w}^*)^{(0)}}(X_i)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\
&\leq \left(4 \cdot \sum_{k=1}^{K_n} |(\mathbf{w}^*)_{1,1,k}^{(L)}|^2 \cdot c_{74} \cdot (\log n)^{4L} \cdot \|((\mathbf{w}^*)_{k,i,j}^{(l)})_{k,i,j,l:l < L} - ((\mathbf{w}^*)^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:l < L}\|^2 \right. \\
&\quad \left. + 16 \cdot \sum_{i=1}^n f_{(\mathbf{w}^*)^{(0)}}(X_i)^2 + \frac{8}{n} \sum_{i=1}^n Y_i^2 \right)^{1/2} \\
&\leq \left(c_{75} \cdot M_n^2 \cdot (\log n)^{4L+1} + 16 \cdot \beta_n^2 + 8 \cdot \beta_n \right)^{1/2}
\end{aligned}$$

$$\leq c_{76} \cdot M_n \cdot (\log n)^{2L+1}.$$

Summarizing these steps yields

$$\begin{aligned} & |F_n(\mathbf{w}^*) - F_n((\mathbf{w}^*)^{(0)})| \\ & \leq c_{77} \cdot (\log n)^{4L+1} \cdot M_n \cdot \|((\mathbf{w}^*)_{1,1,k}^{(L)})_{k=1,\dots,K_n}\| \|(\tilde{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l:l < L} - ((\tilde{\mathbf{w}}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:l < L}\| \end{aligned}$$

Thus (17) is satisfied with

$$D_n = c_{77} \cdot M_n \cdot (\log n)^{4L+1}.$$

Let $\epsilon > 0$ be arbitrary. In the *fifth step of the proof* we show that

$$\mathbf{E}T_{2,n} \leq c_{78} \cdot \frac{n^{\tau \cdot d + \epsilon}}{n}.$$

Let $\mathcal{W}_n$ be the set of all weight vectors $(w_{k,i,j}^{(l)})_{k,i,j,l}$ which satisfy

$$|w_{1,1,k}^{(L)}| \leq c_{79} \cdot (\log n)^2 \quad (k = 1, \dots, K_n),$$

$$|w_{k,i,j}^{(l)}| \leq (20d + 1) \cdot (\log n)^2 \quad (l = 1, \dots, L - 1)$$

and

$$|w_{k,i,j}^{(0)}| \leq (8d + 1) \cdot (\log n)^2 \cdot n^\tau.$$

From Lemma 5.1 we know that for sufficiently large n it holds that

$$\|((\mathbf{w}^{(t_n)})_{1,1,k}^{(L)})_{k=1,\dots,K_n} - ((\mathbf{w}^{(0)})_{1,1,k}^{(L)})_{k=1,\dots,K_n}\| \leq \sqrt{2F_n(\mathbf{w}^{(0)})} \leq (\log n)^2$$

and

$$\|((\mathbf{w}^{(t_n)})_{k,i,j}^{(l)})_{k,i,j,l:l < L} - ((\mathbf{w}^{(0)})_{k,i,j}^{(l)})_{k,i,j,l:l < L}\| \leq \sqrt{2F_n(\mathbf{w}^{(0)})} \leq (\log n)^2.$$

This, together with the initial choice of $\mathbf{w}^{(0)}$ imply that we have

$$\mathbf{w}^{(t)} \in \mathcal{W}_n \quad (t = 0, \dots, t_n)$$

on A_n for sufficiently large n .

Thus, for any $u > 0$ we get

$$\begin{aligned} & \mathbf{P}\{T_{2,n} > u\} \\ & \leq \mathbf{P}\left\{\exists f \in \mathcal{F}_n : \mathbf{E}\left(\left|\frac{f(X)}{\beta_n} - \frac{T_{\beta_n}Y}{\beta_n}\right|^2\right) - \mathbf{E}\left(\left|\frac{m_{\beta_n}(X)}{\beta_n} - \frac{T_{\beta_n}Y}{\beta_n}\right|^2\right) \right. \\ & \quad \left. - \frac{1}{n} \sum_{i=1}^n \left(\left|\frac{f(X_i)}{\beta_n} - \frac{T_{\beta_n}Y_i}{\beta_n}\right|^2 - \left|\frac{m_{\beta_n}(X_i)}{\beta_n} - \frac{T_{\beta_n}Y_i}{\beta_n}\right|^2\right)\right\} \\ & > \frac{1}{2} \cdot \left(\frac{u}{\beta_n^2} + \mathbf{E}\left(\left|\frac{f(X)}{\beta_n} - \frac{T_{\beta_n}Y}{\beta_n}\right|^2\right) - \mathbf{E}\left(\left|\frac{m_{\beta_n}(X)}{\beta_n} - \frac{T_{\beta_n}Y}{\beta_n}\right|^2\right)\right), \end{aligned}$$

where

$$\mathcal{F}_n = \{T_{\beta_n}f_{\mathbf{w}} : \mathbf{w} \in \mathcal{W}_n\}.$$

Applying Lemma 5.4 for $x_1^n \in (\text{supp}(X))^n$ and $\alpha = c_{80}$ we get

$$\begin{aligned} \mathcal{N}_1 \left(\delta, \left\{ \frac{1}{\beta_n} \cdot f : f \in \mathcal{F}_n \right\}, x_1^n \right) &\leq \mathcal{N}_1 (\delta \cdot \beta_n, \mathcal{F}_n, x_1^n) \\ &\leq \left(\frac{c_{81} \cdot \beta_n}{\delta \cdot \beta_n} \right)^{c_{82} \cdot (\log n)^{2d} \cdot n^{\tau \cdot d} \cdot (\log n)^{2 \cdot (L-1) \cdot d} \cdot \left(\frac{\kappa_n \cdot (\log n)^2}{\delta \cdot \beta_n} \right)^{d/k} + c_{83}}. \end{aligned}$$

For $\delta > 1/n$ and k large enough, we obtain

$$\mathcal{N}_1 \left(\delta, \left\{ \frac{1}{\beta_n} \cdot f : f \in \mathcal{F}_n \right\}, x_1^n \right) \leq c_{84} \cdot n^{c_{85} \cdot n^{\tau \cdot d + \epsilon/2}}.$$

This together with Theorem 11.4 in Györfi et al. (2002) leads for $u \geq 1/n$ to

$$\mathbf{P}\{T_{2,n} > u\} \leq 14 \cdot c_{84} \cdot n^{c_{85} \cdot n^{\tau \cdot d + \epsilon/2}} \cdot \exp \left(-\frac{n}{5136 \cdot \beta_n^2} \cdot u \right).$$

For $\epsilon_n \geq 1/n$ we can conclude

$$\begin{aligned} \mathbf{E}\{T_{2,n}\} &\leq \epsilon_n + \int_{\epsilon_n}^{\infty} \mathbf{P}\{T_{2,n} > u\} du \\ &\leq \epsilon_n + 14 \cdot c_{84} \cdot n^{c_{85} \cdot n^{\tau \cdot d + \epsilon/2}} \cdot \exp \left(-\frac{n}{5136 \cdot \beta_n^2} \cdot \epsilon_n \right) \cdot \frac{5136 \cdot \beta_n^2}{n}. \end{aligned}$$

Setting

$$\epsilon_n = \frac{5136 \cdot \beta_n^2}{n} \cdot c_{85} \cdot n^{\tau \cdot d + \epsilon/2} \cdot \log n$$

yields the assertion of the fifth step of the proof.

In the *sixth step of the proof* we show

$$\begin{aligned} \mathbf{E}\{T_{4,n}\} &\leq c_{86} \cdot \left(\sup_{\substack{(\bar{w}_{k,i,j}^{(l)})_{k,i,j,l} \\ |\bar{w}_{k,i,j}^{(l)} - w_{k,i,j}^{(l)}| \leq \log n \ (l=0,\dots,L-1)}} \int \left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} \cdot f_{\bar{w},k,1}^{(L)}(x) - m(x) \right|^2 \mathbf{P}_X(dx) \right. \\ &\quad \left. + \frac{\log n}{n} + \frac{n^{\tau \cdot d + \epsilon}}{n} + M_n^2 \cdot (\log n)^{4L+3/2} \right). \end{aligned}$$

Since

$$|T_{\beta_n} z - y| \leq |z - y| \quad \text{for } |y| \leq \beta_n$$

we obtain

$$\begin{aligned} T_{4,n}/2 &= \left[\frac{1}{n} \sum_{i=1}^n |m_n(X_i) - Y_i|^2 - \frac{1}{n} \sum_{i=1}^n |m(X_i) - Y_i|^2 \right] \cdot 1_{A_n} \\ &\leq \left[\frac{1}{n} \sum_{i=1}^n |f_{\bar{w}(t_n)}(X_i) - Y_i|^2 - \frac{1}{n} \sum_{i=1}^n |m(X_i) - Y_i|^2 \right] \cdot 1_{A_n} \end{aligned}$$

$$= \left[F_n(\mathbf{w}^{(t_n)}) - \frac{1}{n} \sum_{i=1}^n |m(X_i) - Y_i|^2 \right] \cdot 1_{A_n}.$$

The application of Lemma 5.1 implies

$$\begin{aligned} & \mathbf{E}\{T_{4,n}/2\} \\ & \leq \mathbf{E}\left\{ \left[F_n(\mathbf{w}^{(0)}) + D_n \cdot \|((\mathbf{w}^*)^{(L)})_{1,1,k}\|_{k=1,\dots,K_n} \cdot \sqrt{2 \cdot F_n(\mathbf{w}^{(0)})} \right. \right. \\ & \quad \left. \left. + \frac{\|((\mathbf{w}^*)^{(L)})_{1,1,k}\|_{k=1,\dots,K_n} - \|((\mathbf{w}^{(0)})^{(L)})_{1,1,k}\|_{k=1,\dots,K_n}\|^2}{2} + \frac{F_n(\mathbf{w}^{(0)})}{t_n} \right. \right. \\ & \quad \left. \left. - \frac{1}{n} \sum_{i=1}^n |m(X_i) - Y_i|^2 \right] \cdot 1_{A_n} \right\} \\ & \leq 2 \cdot \left(\sup_{\substack{(\tilde{\mathbf{w}}_{k,i,j}^{(L)})_{k,i,j,l}: \\ |\tilde{\mathbf{w}}_{k,i,j}^{(L)} - \mathbf{w}_{k,i,j}^{(L)}| \leq \log n \ (l=0,\dots,L-1)}} \int \left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} \cdot f_{\tilde{\mathbf{w}},k,1}^{(L)}(x) - m(x) \right|^2 \mathbf{P}_X(dx) \right) \\ & \quad + \mathbf{E}\left\{ \left[F_n(\mathbf{w}^{(0)}) + D_n \cdot \|((\mathbf{w}^*)^{(L)})_{1,1,k}\|_{k=1,\dots,K_n} \cdot \sqrt{2 \cdot F_n(\mathbf{w}^{(0)})} \right. \right. \\ & \quad \left. \left. + \frac{\|((\mathbf{w}^*)^{(L)})_{1,1,k}\|_{k=1,\dots,K_n} - \|((\mathbf{w}^{(0)})^{(L)})_{1,1,k}\|_{k=1,\dots,K_n}\|^2}{2} + \frac{F_n(\mathbf{w}^{(0)})}{t_n} \right. \right. \\ & \quad \left. \left. - \frac{1}{n} \sum_{i=1}^n |m(X_i) - Y_i|^2 \right. \right. \\ & \quad \left. \left. - 2 \cdot \left(\mathbf{E}\left\{ \left| \sum_{k=1}^{K_n} (\mathbf{w}^*)_{1,1,k}^{(L)} \cdot f_{(\mathbf{w}^*),j,1}^{(L)}(X) - Y \right|^2 \middle| \mathcal{D}_n, \mathbf{w}^{(0)} \right\} - \mathbf{E}\{|m(X) - Y|^2\} \right) \right] \cdot 1_{A_n} \right\}. \end{aligned}$$

Due to step 4 we have

$$D_n = c_{74} \cdot M_n \cdot (\log n)^{4L+1}.$$

Using the same arguments as in step 2 and 5 of the proof we obtain

$$\begin{aligned} & \mathbf{E}\left\{ \left[F_n((\mathbf{w}^*)^{(0)}) - \frac{1}{n} \sum_{i=1}^n |m(X_i) - Y_i|^2 \right. \right. \\ & \quad \left. \left. + c_{87} \cdot M_n \cdot (\log n)^{4L+1} \cdot \|((\mathbf{w}^*)^{(L)})_{1,1,k}\|_{k=1,\dots,K_n} \cdot \sqrt{2 \cdot F_n(\mathbf{w}^{(0)})} \right. \right. \\ & \quad \left. \left. + \frac{\|((\mathbf{w}^*)^{(L)})_{1,1,k}\|_{k=1,\dots,K_n} - \|((\mathbf{w}^{(0)})^{(L)})_{1,1,k}\|_{k=1,\dots,K_n}\|^2}{2} + \frac{F_n(\mathbf{w}^{(0)})}{t_n} \right. \right. \\ & \quad \left. \left. - 2 \cdot \left(\mathbf{E}\left\{ \left| \sum_{k=1}^{K_n} (\mathbf{w}^*)_{1,1,k}^{(L)} \cdot f_{(\mathbf{w}^*),j,1}^{(L)}(X) - Y \right|^2 \middle| \mathcal{D}_n, \mathbf{w}^{(0)} \right\} - \mathbf{E}\{|m(X) - Y|^2\} \right) \right] \cdot 1_{A_n} \right\} \\ & \leq c_{88} \cdot \frac{\log n}{n} + c_{89} \cdot M_n^2 \cdot (\log n)^{4L+3/2} + \frac{M_n^2}{2} + \frac{c_1 \cdot \log n}{t_n} + c_{90} \cdot \frac{n^{\tau \cdot d + \epsilon}}{n}. \end{aligned}$$

This implies

$$\begin{aligned} & \mathbf{E}\{T_{4,n}/2\} \\ & \leq 2 \cdot \left(\sup_{\substack{(\bar{w}_{k,i,j}^{(l)})_{k,i,j,l}: \\ |\bar{w}_{k,i,j}^{(l)} - w_{k,i,j}^{(l)}| \leq \log n \ (l=0,\dots,L-1)}} \int \left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} \cdot f_{\bar{\mathbf{w}},k,1}^{(L)}(x) - m(x) \right|^2 \mathbf{P}_X(dx) \right) \\ & \quad + c_{88} \cdot \frac{\log n}{n} + c_{90} \cdot \frac{n^{\tau \cdot d + \epsilon}}{n} + c_{91} \cdot M_n^2 \cdot (\log n)^{4L+3/2}. \end{aligned}$$

Summarizing the above results we get the assertion. $\square$

To prove Theorem 3.2, we use the following lemma, which provides another bound on the approximation error. Furthermore, it ensures that the outer weights are sufficiently small.

Lemma 5.7. *Let $1/2 \leq p \leq 1$, $C > 0$, let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a (p, C) -smooth function and let X be a $\mathbb{R}^d$ -valued random variable with $\text{supp}(X) \subseteq [0, 1]^d$. Let $l \in \mathbb{N}$, $0 < \delta < 1/2$ with*

$$c_{92} \cdot \delta \leq \frac{1}{2^l} \leq c_{93} \cdot \delta \quad (56)$$

and let $L, r, s \in \mathbb{N}$ with $L \geq 2$ and $r \geq 2d$. Furthermore, let

$$\tilde{K}_n \geq \left(l \cdot (2^l + 1)^{2d} + 1 \right)^3.$$

Then there exist

$$w_{1,1,k}^{(L)} \in \mathbb{R} \quad \text{for } k = 1, \dots, \tilde{K}_n,$$

$$w_{k,i,j}^{(l)} \in [-20d \cdot (\log n)^2, 20d \cdot (\log n)^2] \quad \text{for } l = 1, \dots, L-1, k = 1, \dots, \tilde{K}_n$$

and

$$w_{k,i,j}^{(0)} \in \left[-\frac{8 \cdot d \cdot (\log n)^2}{\delta}, \frac{8 \cdot d \cdot (\log n)^2}{\delta} \right] \quad \text{for } k = 1, \dots, \tilde{K}_n$$

such that for all $\bar{\mathbf{w}}$ satisfying $|\bar{w}_{k,i,j}^{(l)} - w_{k,i,j}^{(l)}| \leq \log n$ ($l = 0, \dots, L-1$) we have for n sufficiently large

$$\begin{aligned} & \int \left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} \cdot f_{\bar{\mathbf{w}},k,1}^{(L)}(x) - f(x) \right|^2 \mathbf{P}_X(dx) \\ & \leq c_{94} \cdot \left(l^2 \cdot \delta + \delta^{2p} + \frac{l \cdot (2^l + 1)^{2d}}{n^s} \right), \end{aligned} \quad (57)$$

$$\left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} \cdot f_{\bar{\mathbf{w}},k,1}^{(L)}(x) \right| \leq c_{95} \cdot \left(1 + \frac{(2^l + 1)^{2d}}{n^s} \right) \quad (x \in [0, 1]^d) \quad (58)$$

and

$$\sum_{k=1}^{\tilde{K}_n} |w_{1,1,k}^{(L)}|^2 \leq \frac{c_{96}}{2^{2 \cdot d \cdot l}}. \quad (59)$$

Proof. The proof follows from the proof of Lemma 7 in [Kohler and Krzyżak \(2022b\)](#). $\square$

5.4. Proof of Theorem 3.2

Proof. Let $l = \lfloor \frac{1}{1+d} \log_2 n \rfloor$. Then condition (56) holds for $\delta = n^{-1/(1+d)}$. Set

$$\tilde{K}_n = (l \cdot (2^l + 1)^{2d} + 1)^3 \approx c_{97} \cdot (\log n)^3 \cdot n^{\frac{6d}{1+d}}, \quad N_n = n^{c_{98}}$$

and define the weight vector $\mathbf{w}$ as in Lemma 5.7. Then we obtain by Lemma 5.7 that assumption (49) is satisfied for $M_n = \frac{c_{99}}{n^{d/(1+d)}}$. Assumption (50) follows directly from (58) of Lemma 5.7 for s sufficiently large.

By Lemma 5.6, $\tau = \frac{1}{1+d}$ and Lemma 5.7 where s is sufficiently large, we get for large n

$$\begin{aligned} \mathbf{E} \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) &\leq c_{100} \cdot \left(\frac{n^{\frac{1}{1+d} \cdot d + \epsilon}}{n} + c_{101} \cdot \frac{(\log n)^{4L+3/2}}{n^{\frac{2d}{d+1}}} \right. \\ &\quad \left. + \sup_{\substack{(\tilde{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l} \\ |\tilde{\mathbf{w}}_{k,i,j}^{(l)} - \mathbf{w}_{k,i,j}^{(l)}| \leq \log n \ (l=0, \dots, L-1)}} \int \left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} \cdot f_{\tilde{\mathbf{w}},k,1}^{(L)}(x) - m(x) \right|^2 \mathbf{P}_X(dx) \right) \\ &\leq c_{102} \cdot \left(\frac{n^{\frac{1}{1+d} \cdot d + \epsilon}}{n} + \frac{(\log n)^{4L+3/2}}{n^{\frac{2d}{d+1}}} + \frac{(\log n)^2}{n^{\frac{1}{1+d}}} + \frac{1}{n^{\frac{2p}{1+d}}} + \frac{\log n \cdot n^{\frac{2d}{1+d}}}{n^s} \right) \\ &\leq c_{103} \cdot n^{-\frac{1}{1+d} + \epsilon} \end{aligned}$$

for s sufficiently large. $\square$

5.5. Auxiliary results for the proof of Theorem 3.3

In order to prove Theorem 3.3, we use the following lemma, which controls the complexity of a set of over-parametrized deep neural networks for interaction models.

Lemma 5.8. *Let $\alpha \geq 1$, $\beta > 0$ and let $A, B, C \geq 1$. Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be k -times differentiable such that all derivatives up to order k are bounded on $\mathbb{R}$. Let $\mathcal{F}$ be the set of all functions*

$$f_{\mathbf{w}}(x) = \sum_{I \subseteq \{1, \dots, d\} : |I|=d^*} f_{\mathbf{w}_I}(x_I)$$

where $f_{\mathbf{w}_I}$ is defined by (1)–(3) with d replaced by d^* and weight vector $\mathbf{w}_I$,

$$\mathbf{w} = (\mathbf{w}_I)_{I \subseteq \{1, \dots, d\} : |I|=d^*},$$

and where for any $I \subseteq \{1, \dots, d\}$ with $|I| = d^*$ the weight vector $\mathbf{w}_I$ satisfies

$$\sum_{j=1}^{K_n} |(\mathbf{w}_I)_{1,1,j}^{(L)}| \leq C, \quad (60)$$

$$|(\mathbf{w}_I)_{k,i,j}^{(l)}| \leq B \quad (k \in \{1, \dots, K_n\}, i, j \in \{1, \dots, r\}, l \in \{1, \dots, L-1\}) \quad (61)$$

and

$$|(\mathbf{w}_I)_{k,i,j}^{(0)}| \leq A \quad (k \in \{1, \dots, K_n\}, i \in \{1, \dots, r\}, j \in \{1, \dots, d\}). \quad (62)$$

Then we have for any $1 \leq p < \infty$, $0 < \epsilon < \beta$ and $x_1^n \in [-\alpha, \alpha]^{d \cdot n}$

$$\begin{aligned} & \mathcal{N}_p(\epsilon, \{T_\beta f : f \in \mathcal{F}\}, x_1^n) \\ & \leq \left(c_{104} \cdot \frac{\beta^p}{\epsilon^p} \right)^{c_{105} \cdot \alpha^{d^*} \cdot A^{d^*} \cdot B^{(L-1) \cdot d^*} \left(\frac{\epsilon}{\beta} \right)^{d^*/k} + c_{106}}. \end{aligned}$$

Proof. See Lemma 8 in [Kohler and Krzyżak \(2022b\)](#). $\square$

5.6. Proof of Theorem 3.3

Proof. Let $l = \lfloor \frac{1}{1+d^*} \log_2 n \rfloor$ and $N_n = n^{c_{107}}$. Furthermore, let

$$\tilde{K}_n = \binom{d}{d^*} (l \cdot (2^l + 1)^{2d^*} + 1)^3 \approx c_{108} \cdot (\log n)^3 \cdot n^{\frac{6d^*}{1+d^*}}.$$

Then (56) holds for $\delta = n^{-1/(1+d^*)}$. Define the weight vector $\mathbf{w}$ such that the components are chosen according to Lemma 5.7 so that they approximate m_I .

Assumption (49) and (50) of Theorem 5.6 are satisfied for $\mathbf{w}$ since

$$\sum_{I \subseteq \{1, \dots, d\} : |I|=d^*} \sum_{k=1}^{\tilde{K}_n} |(\mathbf{w}_I)^{(L)}_{1,1,k}|^2 \leq c_{109} \cdot n^{-\frac{2d^*}{d^*+1}}$$

and

$$\sum_{I \subseteq \{1, \dots, d\} : |I|=d^*} \left| \sum_{k=1}^{\tilde{K}_n} (\mathbf{w}_I)^{(L)}_{1,1,k} \cdot f_{(\tilde{\mathbf{w}}_I)_{k,1}}^{(L)}(x) \right| \leq c_{110} \cdot \left(1 + \frac{n^{2d^*/(1+d^*)}}{n^s} \right)$$

hold.

Then we obtain by application of Lemma 5.8 the assertion of Theorem 5.6 for interaction models. Therefore the proof of Theorem 3.3 follows similarly to the proof of Theorem 3.2. By applying the assertions of Theorem 5.6 and Lemma 5.7 we get for $\tau = \frac{1}{1+d^*}$ and s sufficiently large

$$\begin{aligned} \mathbf{E} \int |m_n(x) - m(x)|^2 \mathbf{P}_X(dx) & \leq c_{111} \cdot \left(\frac{n^{\frac{1}{1+d^*} \cdot d^* + \epsilon}}{n} + c_{110} \cdot (\log n)^{4L+3/2} \cdot n^{-\frac{2d^*}{d^*+1}} \right. \\ & \quad + \sup_{\substack{(\tilde{\mathbf{w}}_{k,i,j}^{(l)})_{k,i,j,l} : \\ |\tilde{\mathbf{w}}_{k,i,j}^{(l)} - \mathbf{w}_{k,i,j}^{(l)}| \leq \log n \ (l=0, \dots, L-1)}} \int \left| \sum_{k=1}^{\tilde{K}_n} w_{1,1,k}^{(L)} \cdot f_{\tilde{\mathbf{w}},k,1}^{(L)}(x) - m(x) \right|^2 \mathbf{P}_X(dx) \Big) \\ & \leq c_{112} \cdot \left(\frac{n^{\frac{1}{1+d^*} \cdot d^* + \epsilon}}{n} + \frac{(\log n)^{4L+3/2}}{n^{\frac{2d^*}{1+d^*}}} + \frac{(\log n)^2}{n^{\frac{1}{1+d^*}}} + \frac{1}{n^{\frac{2p}{1+d^*}}} + \frac{\log n \cdot n^{\frac{2d^*}{d^*+1}}}{n^s} \right) \\ & \leq c_{113} \cdot n^{-\frac{1}{1+d^*} + \epsilon}. \end{aligned}$$

$\square$

Appendix A: Detailed Proofs of Lemma 5.2 and Lemma 5.3**A.1. Proof of Lemma 5.2**

We have

$$\begin{aligned}
 \|\nabla_{\mathbf{w}} F_n(\mathbf{w})\|^2 &= \sum_{k,i,j,l} \left(\frac{2}{n} \sum_{s=1}^n (Y_s - f_{\mathbf{w}}(X_s)) \cdot \frac{\partial f_{\mathbf{w}}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2 \\
 &\leq 4 \cdot \sum_{k,i,j,l} \frac{1}{n} \sum_{s=1}^n (Y_s - f_{\mathbf{w}}(X_s))^2 \cdot \frac{1}{n} \sum_{s=1}^n \left(\frac{\partial f_{\mathbf{w}}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2 \\
 &\leq c_{114} \cdot K_n \cdot L \cdot r^2 \cdot d \cdot \max_{k,i,j,l,s} \left(\frac{\partial f_{\mathbf{w}}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2 \cdot \frac{1}{n} \sum_{s=1}^n (Y_s - f_{\mathbf{w}}(X_s))^2.
 \end{aligned}$$

The partial derivative of f is given by

$$\begin{aligned}
 \frac{\partial f_{\mathbf{w}}}{\partial w_{k,i,j}^{(l)}}(x) &= \sum_{s_{l+2}=1}^r \cdots \sum_{s_{L-1}=1}^r f_{k,j}^{(l)}(x) \cdot \sigma' \left(\sum_{t=1}^r w_{k,i,t}^{(l)} \cdot f_{k,t}^{(l)}(x) + w_{k,i,0}^{(l)} \right) \\
 &\quad \cdot w_{k,s_{l+2},i}^{(l+1)} \cdot \sigma' \left(\sum_{t=1}^r w_{k,s_{l+2},t}^{(l+1)} \cdot f_{k,t}^{(l+1)}(x) + w_{k,s_{l+2},0}^{(l+1)} \right) \cdot w_{k,s_{l+3},s_{l+2}}^{(l+2)} \\
 &\quad \cdot \sigma' \left(\sum_{t=1}^r w_{k,s_{l+3},t}^{(l+2)} \cdot f_{k,t}^{(l+2)}(x) + w_{k,s_{l+3},0}^{(l+2)} \right) \cdots w_{k,s_{L-1},s_{L-2}}^{(L-2)} \\
 &\quad \cdot \sigma' \left(\sum_{t=1}^r w_{k,s_{L-1},t}^{(L-2)} \cdot f_{k,t}^{(L-2)}(x) + w_{k,s_{L-1},0}^{(L-2)} \right) \cdot w_{k,1,s_{L-1}}^{(L-1)} \\
 &\quad \cdot \sigma' \left(\sum_{t=1}^r w_{k,1,t}^{(L-1)} \cdot f_{k,t}^{(L-1)}(x) + w_{k,1,0}^{(L-1)} \right) \cdot w_{1,1,k}^{(L)}, \tag{63}
 \end{aligned}$$

where we have used the abbreviations

$$f_{k,j}^{(0)}(x) = \begin{cases} x^{(j)} & \text{if } j \in \{1, \dots, d\} \\ 1 & \text{if } j = 0 \end{cases}$$

and

$$f_{k,0}^{(l)}(x) = 1 \quad (l = 1, \dots, L-1).$$

Together with assumptions (21) and (22) we obtain

$$\max_{k,i,j,l,s} \left(\frac{\partial f_{\mathbf{w}}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2 \leq c_{115} \cdot r^{2L} \cdot \max\{\|\sigma'\|_{\infty}^{2L}, 1\} \cdot B_n^{2L} \cdot (\gamma_n^*)^2 \cdot \alpha_n^2.$$

In the next step of the proof we want to show, that

$$|f_{\mathbf{w}}(x) - f_{\mathbf{v}}(x)| \leq 2 \cdot K_n \cdot \max\{\|\sigma'\|_{\infty}^L, 1\} \cdot \gamma_n^* \cdot (2r+1)^L \cdot B_n^L \cdot \alpha_n \cdot \|\mathbf{w} - \mathbf{v}\|_{\infty} \cdot \max\{\|\sigma\|_{\infty}, 1\}.$$

Let $\bar{f}_{k,i}^{(l)}$ be defined by

$$\bar{f}_{k,i}^{(l)}(x) = \sigma \left(\sum_{j=1}^r v_{k,i,j}^{(l-1)} \cdot \bar{f}_{k,j}^{(l-1)}(x) + v_{k,i,0}^{(l-1)} \right)$$

for $l = 2, \dots, L$ and

$$\bar{f}_{k,i}^{(1)}(x) = \sigma \left(\sum_{j=1}^d v_{k,i,j}^{(0)} \cdot x^{(j)} + v_{k,i,0}^{(0)} \right).$$

First, we show by induction that

$$\begin{aligned} |f_{k,i}^{(l)}(x) - \bar{f}_{k,i}^{(l)}(x)| &\leq \max\{\|\sigma'\|_\infty^l, 1\} \cdot (2r+1)^l \cdot B_n^l \cdot \alpha_n \\ &\quad \cdot \max_{i,j,s:s < L} |w_{k,i,j}^{(s)} - v_{k,i,j}^{(s)}| \cdot \max\{\|\sigma\|_\infty, 1\} \end{aligned} \quad (64)$$

holds for $l = 1, \dots, L$ and $x \in [-\alpha_n, \alpha_n]^d$.

The function σ is differentiable and its derivative is bounded, hence σ is Lipschitz continuous with Lipschitz constant $\|\sigma'\|_\infty$. This implies

$$\begin{aligned} |f_{k,i}^{(1)}(x) - \bar{f}_{k,i}^{(1)}(x)| &\leq \|\sigma'\|_\infty \cdot \left(\sum_{j=1}^d |w_{k,i,j}^{(0)} - v_{k,i,j}^{(0)}| \cdot |x^{(j)}| + |w_{k,i,0}^{(0)} - v_{k,i,0}^{(0)}| \right) \\ &\leq \|\sigma'\|_\infty \cdot (2r+1) \cdot \alpha_n \cdot \max_{i,j,s:s < L} |w_{k,i,j}^{(s)} - v_{k,i,j}^{(s)}|. \end{aligned}$$

Assume (64) holds for some $l-1$ with $l = 2, \dots, L-1$. Then we have

$$\begin{aligned} &|f_{k,i}^{(l)}(x) - \bar{f}_{k,i}^{(l)}(x)| \\ &\leq \|\sigma'\|_\infty \cdot \left(\sum_{j=1}^r |w_{k,i,j}^{(l-1)}| \cdot |f_{k,j}^{(l-1)}(x) - \bar{f}_{k,j}^{(l-1)}(x)| \right. \\ &\quad \left. + \sum_{j=1}^r |w_{k,i,j}^{(l-1)} - v_{k,i,j}^{(l-1)}| \cdot |\bar{f}_{k,j}^{(l-1)}(x)| + |w_{k,i,0}^{(l-1)} - v_{k,i,0}^{(l-1)}| \right) \\ &\leq \|\sigma'\|_\infty \cdot \left(r \cdot B_n \cdot \max_{j=1,\dots,r} |f_{k,j}^{(l-1)}(x) - \bar{f}_{k,j}^{(l-1)}(x)| \right. \\ &\quad \left. + (r+1) \cdot \max_{i,j,s:s < L} |w_{k,i,j}^{(s)} - v_{k,i,j}^{(s)}| \cdot \max\{\|\sigma\|_\infty, 1\} \right) \\ &\leq \max\{\|\sigma'\|_\infty^l, 1\} \cdot (2r+1)^l \cdot B_n^l \cdot \alpha_n \cdot \max_{i,j,s:s < L} |w_{k,i,j}^{(s)} - v_{k,i,j}^{(s)}| \cdot \max\{\|\sigma\|_\infty, 1\}. \end{aligned}$$

This implies

$$\begin{aligned} &|f_{\mathbf{w}}(x) - f_{\mathbf{v}}(x)| \\ &= \left| \sum_{k=1}^{K_n} w_{1,1,k}^{(L)} \cdot f_{k,1}^{(L)}(x) - \sum_{k=1}^{K_n} v_{1,1,k}^{(L)} \cdot \bar{f}_{k,1}^{(L)}(x) \right| \end{aligned}$$

$$\begin{aligned}
& \leq \left| \sum_{k=1}^{K_n} w_{1,1,k}^{(L)} \left(f_{k,1}^{(L)}(x) - \bar{f}_{k,1}^{(L)}(x) \right) \right| + \left| \sum_{k=1}^{K_n} \left(w_{1,1,k}^{(L)} - v_{1,1,k}^{(L)} \right) \cdot \bar{f}_{k,1}^{(L)}(x) \right| \\
& \leq K_n \cdot \max_k \left| w_{1,1,k}^{(L)} \right| \cdot \max_k \left| f_{k,1}^{(L)}(x) - \bar{f}_{k,1}^{(L)}(x) \right| \\
& \quad + K_n \cdot \max_k \left| w_{1,1,k}^{(L)} - v_{1,1,k}^{(L)} \right| \cdot \max\{\|\sigma\|_\infty, 1\} \\
& \leq 2 \cdot K_n \cdot \gamma_n^* \cdot \max\{\|\sigma'\|_\infty^L, 1\} \cdot (2r+1)^L \cdot B_n^L \cdot \alpha_n \cdot \|\mathbf{w} - \mathbf{v}\|_\infty \cdot \max\{\|\sigma\|_\infty, 1\}.
\end{aligned}$$

Together with assumption (23) we can conclude

$$\begin{aligned}
& \frac{1}{n} \sum_{s=1}^n (Y_s - f_{\mathbf{w}}(X_s))^2 \\
& \leq 2 \cdot F_n(\mathbf{v}) + \frac{2}{n} \sum_{s=1}^n (f_{\mathbf{v}}(X_s) - f_{\mathbf{w}}(X_s))^2 \\
& \leq 2 \cdot F_n(\mathbf{v}) + 8 \cdot K_n^2 \cdot (\gamma_n^*)^2 \cdot \max\{\|\sigma'\|_\infty^{2L}, 1\} \cdot (2r+1)^{2L} \cdot B_n^{2L} \cdot \alpha_n^2 \\
& \quad \cdot \max\{\|\sigma\|_\infty, 1\}^2 \cdot \frac{8t_n}{L_n} \cdot \max\{F_n(\mathbf{v}), 1\}.
\end{aligned}$$

By summarizing the above results we obtain the assertion. $\square$

A.2. Proof of Lemma 5.3

We have

$$\begin{aligned}
& \|\nabla_{\mathbf{w}} F_n(\mathbf{w}_1) - \nabla_{\mathbf{w}} F_n(\mathbf{w}_2)\|^2 \\
& = \sum_{k,i,j,l} \left(\frac{2}{n} \sum_{s=1}^n (Y_s - f_{\mathbf{w}_1}(X_s)) \cdot \frac{\partial f_{\mathbf{w}_1}}{\partial w_{k,i,j}^{(l)}}(X_s) \right. \\
& \quad \left. - \left(\frac{2}{n} \sum_{s=1}^n (Y_s - f_{\mathbf{w}_2}(X_s)) \cdot \frac{\partial f_{\mathbf{w}_2}}{\partial w_{k,i,j}^{(l)}}(X_s) \right) \right)^2 \\
& \leq 8 \cdot \sum_{k,i,j,l} \left(\frac{1}{n} \sum_{s=1}^n (f_{\mathbf{w}_2}(X_s) - f_{\mathbf{w}_1}(X_s)) \cdot \frac{\partial f_{\mathbf{w}_1}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2 \\
& \quad + 8 \cdot \sum_{k,i,j,l} \left(\frac{1}{n} \sum_{s=1}^n (Y_s - f_{\mathbf{w}_2}(X_s)) \cdot \left(\frac{\partial f_{\mathbf{w}_1}}{\partial w_{k,i,j}^{(l)}}(X_s) - \frac{\partial f_{\mathbf{w}_2}}{\partial w_{k,i,j}^{(l)}}(X_s) \right) \right)^2 \\
& \leq 8 \cdot \sum_{k,i,j,l} \max_{s=1,\dots,n} \left(\frac{\partial f_{\mathbf{w}_1}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2 \cdot \frac{1}{n} \sum_{s=1}^n (f_{\mathbf{w}_2}(X_s) - f_{\mathbf{w}_1}(X_s))^2 \\
& \quad + 8 \cdot \frac{1}{n} \sum_{s=1}^n (Y_s - f_{\mathbf{w}_2}(X_s))^2 \cdot \sum_{k,i,j,l} \max_{s=1,\dots,n} \left(\frac{\partial f_{\mathbf{w}_1}}{\partial w_{k,i,j}^{(l)}}(X_s) - \frac{\partial f_{\mathbf{w}_2}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2.
\end{aligned}$$

From the proof of Lemma 5.2 we can conclude

$$\sum_{k,i,j,l} \max_{s=1,\dots,n} \left(\frac{\partial f_{\mathbf{w}_1}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2 \leq c_{116} \cdot K_n \cdot L \cdot r^2 \cdot d \cdot r^{2L} \cdot \max\{\|\sigma'\|_\infty^{2L}, 1\} \cdot B_n^{2L} \cdot (\gamma_n^*)^2 \cdot \alpha_n^2,$$

$$\begin{aligned} & \frac{1}{n} \sum_{s=1}^n (f_{\mathbf{w}_2}(X_s) - f_{\mathbf{w}_1}(X_s))^2 \\ & \leq 4 \cdot K_n^2 \cdot (\gamma_n^*)^2 \cdot \max\{\|\sigma'\|_\infty^{2L}, 1\} \cdot (2r+1)^{2L} \cdot B_n^{2L} \cdot \alpha_n^2 \cdot \max\{\|\sigma\|_\infty, 1\}^2 \|\mathbf{w}_1 - \mathbf{w}_2\|^2, \end{aligned}$$

and

$$\begin{aligned} & \frac{1}{n} \sum_{s=1}^n (Y_s - f_{\mathbf{w}_2}(X_s))^2 \\ & \leq 2 \cdot F_n(\mathbf{v}) + 8 \cdot K_n^2 \cdot (\gamma_n^*)^2 \cdot \max\{\|\sigma'\|_\infty^{2L}, 1\} \cdot (2r+1)^{2L} \cdot B_n^{2L} \cdot \alpha_n^2 \cdot \max\{\|\sigma\|_\infty, 1\}^2 \\ & \quad \frac{8t_n}{L_n} \cdot \max\{F_n(\mathbf{v}), 1\}. \end{aligned}$$

So it remains to bound

$$\sum_{k,i,j,l} \max_{s=1,\dots,n} \left(\frac{\partial f_{\mathbf{w}_1}}{\partial w_{k,i,j}^{(l)}}(X_s) - \frac{\partial f_{\mathbf{w}_2}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2.$$

By (63) we know that

$$\frac{\partial f_{\mathbf{w}}}{\partial w_{k,i,j}^{(l)}}(x)$$

for fixed $x \in [-\alpha_n, \alpha_n]^d$ is a sum of at most r^{L-2} products where each product contains at most $2L+1$ factors. Each of these products contains at most L factors, that are bounded in absolute value by B_n except the last one, which is bounded in absolute value by γ_n^* . Considered as a function in $\mathbf{w}$, these products are Lipschitz continuous with a Lipschitz constant bounded by 1.

According to the proof of Lemma 5.2 we know that $f_{k,j}^{(l)}(x)$, which is either bounded by $\|\sigma\|_\infty$ or α_n , is Lipschitz continuous with a Lipschitz constant bounded by

$$\max\{\|\sigma'\|_\infty^l, 1\} \cdot \max\{\|\sigma\|_\infty, 1\} \cdot (2r+1)^l \cdot B_n^l \cdot \alpha_n.$$

The remaining at most L factors are bounded by $\max\{\|\sigma'\|_\infty, 1\}$ with a Lipschitz constant bounded by

$$c_{117} \cdot (2r+1)^L \cdot B_n^L \cdot \alpha_n \cdot \max\{\|\sigma\|_\infty, 1\}.$$

The assertion follows from the following result: If $g_1, \dots, g_s : \mathbb{R} \rightarrow \mathbb{R}$ are Lipschitz continuous functions with Lipschitz constants $C_{\text{Lip}, g_1}, \dots, C_{\text{Lip}, g_s}$, then

$$\prod_{l=1}^s g_l \quad \text{and} \quad \sum_{l=1}^s g_l$$

are Lipschitz continuous functions with Lipschitz constants bounded by

$$\sum_{l=1}^s C_{Lip, g_l} \cdot \prod_{k \in \{1, \dots, s\} \setminus \{l\}} \|g_k\|_{\infty} \leq s \cdot \max_l C_{Lip, g_l} \cdot \prod_{k \in \{1, \dots, s\} \setminus \{l\}} \|g_k\|_{\infty}$$

and

$$\sum_{l=1}^s C_{Lip, g_l} \leq s \cdot \max_L C_{Lip, g_l}$$

respectively.

This together with the fact that σ and σ' are bounded yields

$$\sum_{k,i,j,l} \max_{s=1, \dots, n} \left(\frac{\partial f_{\mathbf{w}_1}}{\partial w_{k,i,j}^{(l)}}(X_s) - \frac{\partial f_{\mathbf{w}_2}}{\partial w_{k,i,j}^{(l)}}(X_s) \right)^2 \leq c_{118} \cdot K_n \cdot B_n^{4L} \cdot \alpha_n^4 \cdot (\gamma_n^*)^2 \cdot \|\mathbf{w}_1 - \mathbf{w}_2\|^2.$$

Summarizing the above results we get the assertion of Theorem 5.3. $\square$

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